The Dawn of Civilization
Metal Trade and the Rise of Hierarchy*

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Abstract

The Bronze Age witnessed the emergence of the first political states and settlements of unprecedented size across several distant regions of the Old World. The underlying causes of this development are still debated. Drawing on novel archaeological data, we present new empirical evidence suggesting that the spread of metallurgy—and the resulting expansion of long-distance trade—likely played a significant role. We show that regions lying along corridors that linked metal mines to fertile agricultural zones were more likely to develop state institutions and urban centers. We propose that transit bottlenecks along these routes fostered the rise of a new toll-collecting elite, thereby accelerating this development. We formally test this appropriability hypothesis and provide anecdotal evidence based on detailed historical case studies.

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1 Introduction

The second half of the fourth millennium BC marked a turning point in human history: the beginning of the Urban Revolution, a transformation defined by two key developments—the emergence of centralized state polities and an unprecedented expansion in the scale of settlements. Between the fourth and third millennia BC, state-level societies appeared across geographically distant regions, including the Nile Valley, Anatolia, Mesopotamia, Central Asia, and the Indus Valley. By the early second millennium BC, similar transformations had also occurred in China and the Mediterranean Basin. In most cases, state formation was closely tied to the growth of urban centers.

The emergence and spread of the Urban Revolution raises fundamental questions. What caused this transition? Why did it emerge in some regions but not in others? Since the work of moral philosophers such as Thomas Hobbes, John Locke, and Jean-Jacques Rousseau, social scientists have sought to explain the origins of states and political inequality. These questions remain central to the field of comparative development, as an early start in the civilization process has been proposed as a key factor in explaining long-term economic disparities between nations today.[1]

This paper investigates a specific factor that may have played an important role in fostering an earlier development of state societies and urban settlements in the Old World: the rise of long-distance trade. In the fourth millennium BC, copper metallurgy spread across vast regions of the Old World, and bronze was invented. Maintaining a steady supply of metals, however, required complex trade arrangements for two reasons. First, metal resources were scarce in many densely populated areas. Second, bronze is an alloy primarily composed of copper and tin (or, in earlier instances, arsenic)—metals that were often located in geographically distant regions. These constraints catalyzed what has been described as the earliest wave of trade globalization in human history (Vandkilde, 2016). We conjecture that the formation of long-distance metal trade networks was one of the likely contributors to the Urban Revolution.[2]

To test this hypothesis, we take a three-step approach: (1) identify the transit regions in the metal trade network, (2) assess the relationship between metal transit trade and the emergence of the Urban Revolution, and (3) examine a potential mechanism—the rise of an elite class that could levy taxes on metal traders.

To identify the metal transit regions, we draw on a concept introduced by Ramsay (1890), who coined the term "road-knots" to describe locations where multiple routes—shaped by natural topography and the distribution of resources—intersect. We begin by reconstructing the Bronze Age natural transportation network. Using standard methodologies from the trade literature, together with data on topography and ancient bilateral trade relationships, we determine the least-cost paths

[1] Recent research in the field of comparative development has highlighted an exceptional persistence in the levels of economic, technological, and political development around the world until 1500 AD (Comin, Easterly and Gong, 2010; Ashraf and Galor, 2011; Maloney and Valencia Caicedo, 2016; Davis and Weinstein, 2002; Olsson and Paik, 2020). Although the European colonization seems to have created a "reversal of fortune" among European colonies (Acemoglu, Johnson and Robinson, 2002), once accounting for migration between countries, ancient economic disparities are still explaining a large portion of current economic disparities (Putterman and Weil, 2010; Chanda, Cook and Putterman, 2014). See Galor (2011) and Galor (2022) for an exhaustive review of the main theories and narratives behind the long-term persistence of economic outcomes. Matranga and Pascali (2021) provide a review of the empirical work.

[2] Let us emphasize here that complex hierarchies and trade predate the invention of bronze. What is fundamentally different in the Bronze Age is the scale of both hierarchical complexity and long-distance exchange. During the Neolithic, we see chiefs who were able to control the surplus produced in areas with storable crops, but no states. Furthermore exchange networks were regional with occasional short-lived long-distance connections limited to prestige goods. For an exhaustive comparison of Neolithic versus Bronze-Age trade, see Kristiansen (2015), who concludes: "Nowhere during the Neolithic and Copper Age do we find permanently organized, long-distance ('international') trade networks of the kind provided all Bronze Age communities with metal and other wealth from a few source areas on a regular basis".

between all 1×1 degree grid cells across the Old World. We then identify grid cells that contained tin or copper mines during the Bronze Age, as well as those with cropland. Finally, for each grid cell, we compute the number of least-cost paths—linking mines to cropland—that pass through it. This "metal transit index" measures the transit trade potential at each location and serves to identify the road-knots in the metal trade network.

Next, we examine the relationship between this metal transit index and the rise of the Urban Revolution. To our knowledge, there does not exist an ideal global database that coherently measures the defining aspects of the Urban Revolution. We therefore perform multiple imperfect empirical tests based on different datasets.

The first empirical exercise focuses on the emergence of urbanism. Our main dataset for capturing urbanism is Green et al. (2024). In robustness checks we use three additional datasets (Reba, Reitsma and Seto (2016), Lawrence, Hinojosa-Balino and Chelazzi (2025), and Klein Goldewijk et al. (2011)). Figure 1 illustrates that urbanism tended to emerge in potential transit areas of metal trade routes—a pattern confirmed by cross-sectional OLS estimates. Still, several factors prevent a causal interpretation of this correlation. In particular, the metal transit index depends on the distribution of mines and cropland areas around the globe, which are clearly endogenous to the location of cities. To get closer to a causal link, we use a 2SLS estimation framework. The instrumental variable is an alternative measure of transit trade potential, which does not rely on the endogenous distribution of metal mines and cropland, is not affected by human intervention, and is purely driven by geography and climate. It is constructed by counting the number of least-cost paths that transit a certain cell, while connecting the cells with tin and copper natural deposits (not the Bronze Age mines) to the other cells in the Old World, weighted by the maximum potential biomass that can be produced based on the local geo-climatic conditions (a measure of land productivity). The 2SLS regressions confirm the OLS findings. We also observe that while copper transit regions were already significant from the fourth millennium BC, the impact of tin transit regions became prominent only at the very end of the third millennium BC. This likely reflects the transition from arsenic bronze to tin bronze.[3]

The results based on ancient cities have certain limitations. First, due to the cross-sectional nature of the data, we cannot exclude the possibility that omitted geographic factors might be driving the 2SLS results. Evaluating pre-trends is challenging because settlements with populations in the thousands prior to bronze usage (often referred to in archaeology as "megasites") are scarce. In the online appendix, we run a difference-in-differences exercise, indicating that large settlements became more likely in metal transit regions when transitioning from the Neolithic to the Bronze Age. However, this result should be interpreted with caution, as there are only 16 observations in the pre-period. Second, the emergence of urban settlements represents just one dimension of the broader Urban Revolution. While all Bronze Age cities in our sample exhibit state-level organization, not all Bronze Age state societies developed through urbanism. Notably, some centralized state formations—such as Egypt during the Early Dynastic Period (ca. 3100 BC)—emerged without the presence of densely populated urban centers.

To address these limitations, we conduct a second empirical exercise in which we employ two additional data sources that report the locations of radiocarbon-dated archaeological sites in the

[3]Tin bronze appears relatively late compared to arsenic bronze. In Mesopotamia, the number of sites with tin bronze increases sharply around 2600 BC. Similarly, tin bronze becomes more prevalent in the eastern Mediterranean between 2600 and 2200 BC, and in Central Europe after 2200 BC. Rahmstorf (2011) provides a detailed chronology.

Old World—sites that presumably indicate the presence of social and political hierarchies. The first source is a historical atlas of significant archaeological sites (Atlas of World Archaeology, Bahn, 2000), and the second is a gazetteer of ancient history (the Pleiades Project, Bagnall, 2022). Using a difference-in-differences approach, we show that road-knots are associated with a relative increase in archaeological sites during the Bronze Age (as compared to the Paleolithic and the Neolithic). Among these sites, the strongest effects are observed for burials that reflect social hierarchy, whereas we do not find any significant effect on other types of burials. These results are supported by both OLS and 2SLS estimates and cannot be explained by pre-existing trends. Our estimates suggest that metal trade may account for a substantial portion—nearly half—of the observed increase in archaeological site density between the Neolithic and Bronze Age periods. Admittedly, these back-of-the-envelope calculations are crude estimates and should be interpreted with caution as they assume that the rollout of the metallurgical techniques was uniform across the globe. Nevertheless, our estimates suggest that the invention and spread of bronze metallurgy was likely not merely a by-product, but rather a significant contributing factor to the Urban Revolution.

The two historical atlases used in this exercise come with important caveats: one is not an academic publication, and the second one is skewed towards Europe, the Middle East, and Mediterranean Africa. To address these limitations, we zoom in on smaller geographic areas using high-quality local data in the next two empirical exercises.

In the third empirical exercise, we use the Atlas of Chinese Cultural Relics, a 40-volume atlas published by the Chinese National Heritage Administration with detailed information on 500,000+ archaeological sites dating from the Paleolithic onwards. We digitize the atlas and use a naïve Bayesian text algorithm to identify sites indicative of the rise of the Urban Revolution. Specifically, the algorithm recognizes a series of standard archaeological markers of the Urban Revolution. Using both an OLS and 2SLS difference-in-differences design, we find that metal transit regions are associated with a significant increase in all of these markers during the Bronze Age relative to the other regions of China.

In the fourth empirical exercise, we examine Bronze Age Europe. We digitize the Prähistorische Bronzefunde, a collection of monographs on metal artifacts dating from 3000 to 500 BC, which allows us to trace the spread of elite burials. With the exception of the Southeast, which we do not include in our analysis, Europe did not witness the emergence of political states or large cities during the Bronze Age. However, the continent did experience the development of larger polities characterized by a hierarchical settlement structure and unprecedented social inequalities. One of the advantages of focusing on Europe is that we can date the majority of copper and tin mines serving the continent. We show that changes in the road-knots induced precisely by new mineral discoveries in the late Bronze Age correlate with changes in the location of elite burials.

Why were road-knots in the metal trade network likely to contribute to the emergence of complex hierarchies? We propose a simple theoretical model centered on the appropriability of transit trade. The expansion of long-distance exchange created opportunities for a new elite to emerge by taxing goods in transit. In the model, transit trade is appropriable not just by a would-be elite, composed of organized stationary bandits, but also by unorganized roving bandits. The fundamental difference between the two groups is that stationary bandits can commit to a tax rate. Although elites in our model are not necessarily benevolent, the shift from roving to stationary bandits constitutes a Pareto improvement. Commitment benefits both sides: stationary bandits can choose a revenue-

maximizing tax rate, while traders benefit from reduced predation. However, this transition is not costless: it requires coordinating the bandits and securing their monopoly on violence, which entails a fixed cost. Eventually, the shift to stationary appropriation occurs only in the bottlenecks of the metal trade network, where rerouting is prohibitively expensive. The maximum tax revenue that can be extracted from any location is bounded by the cost traders would incur to bypass it; only in bottlenecks do these bounds exceed the fixed costs of establishing stationary banditry. The model can also accommodate a benevolent elite, which uses tax revenues to provide public goods or redistribute them. Thus, it can explain not only the emergence of states characterized by substantial social and economic inequality (e.g., along the Nile Valley) but also the emergence of more egalitarian states (e.g., along the Indus Valley). This appropriability theory borrows heavily from recent works of Timothy Earle and Kristian Kristiansen (Earle (2002, 2013); Earle and Spriggs (2015); Earle et al. (2015); Kristiansen and Suchowska-Ducke (2015); Kristiansen and Earle (2015)). In the context of Europe and the Mediterranean Basin, these authors argue that constriction points in the Bronze Age trade network created the potential for social segments to control the production and flows of critical metal goods. This eventually led to more complex and ranked societies and the rise of networks of integrated polities. For instance, Earle et al. (2015) conjecture that "chiefdoms probably arose at mountain passes, river crossings, and other constriction points, where moving metal and other wealth would have then required payments for safe passage through each local polity".

We test for this appropriability mechanism. Guided by the model, we construct a grid-cell measure of the potential revenues from taxing local transit trade. For each grid cell in the Old World, we calculate the blockage cost—the increase in global transportation costs within the metal trade network that would result from removing that cell. This measure identifies trade bottlenecks—locations where control could yield significant taxing power. We then study whether the Urban Revolution is better predicted by our measure of potential transit trade volume (i.e., the metal transit index) or by our measure of potential fiscal revenues (i.e., the blockage cost). When both variables are included in the regression, only the latter consistently predicts the emergence of cities, settlements, and archaeological sites associated with hierarchical societies during the Bronze Age. This finding supports the appropriability mechanism as a plausible explanation for the empirical connection between transit trade and the Urban Revolution, though several caveats remain.

To provide further evidence in support of the appropriability theory, the last part of the article presents detailed case studies that highlight the role of trade in the rise of political states and urban centers in Mesopotamia, Anatolia, the Indus Valley, the Aegean Sea, and China.

Our work contributes to two separate bodies of literature in the social sciences. First, it contributes to a large empirical literature in economics that studies the role of international connections and market access for comparative development. To our knowledge, our work is a first attempt, within this literature, to go back to the very beginning of the civilization process, has a global scope and provides some empirical validation to the view that trade likely contributed to the development of the first urban civilizations. The closest work is Barjamovic et al. (2019), which shows that natural transportation networks are critical in explaining the hierarchy of ancient city sizes in Central Turkey during the Middle Bronze Age (2000-1650 BC).[4]

[4] A recent literature has documented the role of trade in ancient development during the Iron Age: Bakker et al. (2021) on Phoenician cities, Adamson (2025) on Greek poleis, and Boehm and Chaney (2024) on first-millennium-AD Europe.

Second, our findings contribute to the literature emphasizing the role of geography in early state formation. The conventional wisdom in this literature attributes the rise of complex hierarchies to the spread of agriculture, mainly via three mechanisms: the surplus production enabled by farming made it possible to have an elite that did not work for its own subsistence (productivity theories); the increased vulnerability of farmers to expropriation created both demand for a state that could provide protection and made it possible to sustain a state through a tax system (appropriability theories); and the necessity of public goods such as irrigation infrastructure created a demand for a state (functionalist theories).[5,6,7] These perspectives help explain why the fertile floodplains of the Tigris–Euphrates, Nile, Indus–Ghaggar-Hakra, and Yellow Rivers became early cradles of civilization: they offered abundant yields, easily captured surpluses, and ample scope for irrigation and other large-scale infrastructure. Our work analyzes a complementary mechanism. In the Bronze Age, rivers served as the highways of antiquity and were instrumental to the emergence of a toll-collecting elite—especially at points along the river system that could not be bypassed. In the words of Earle et al. (2015): "The riverine routes channelled metals and other commodities along waterways [...]. In situations where large rivers provided particularly effective, high-volume trade routes, settled groups could have constricted flows and extracted payments for safe passage." One advantage of the proposed mechanism is that it can explain not only riverine cities but also early coastal cities, such as Troy, situated at strategic trade nodes but surrounded by sea and unremarkable farmland. Notice that the long-distance trade explanation of the origin of the Urban Revolution also provides some rationality for a fascinating hypothesis that relates the invention of ironworking with the collapse of the Bronze Age civilizations.[8]

[5]The productivity theories on the rise of early complex hierarchies have a long pedigree in social sciences. Adam Smith argued that the Neolithic transition produced large farming surpluses in some regions. These surpluses made it possible to have a class of bureaucrats, as a result of labor specialization, and increased the demand for an extended role of government (Smith, 1896, p. 65). Friedrich Engels (1884) argued similarly that the surplus generated by the adoption of agriculture was a prerequisite for the transition to a class society. More recently, a growing empirical literature in comparative development has shown that a richer prehistoric availability of domesticates is an important predictor of the timing of the Neolithic transition (Ashraf and Galor, 2011), the early emergence of macro-level polities (Borcan, Olsson and Putterman, 2018; Petersen and Skaaning, 2010), independence from colonial powers post 1500 AD (Ertan, Fiszbein and Putterman, 2016), and higher per-capita GDP today (Bleaney and Dimico, 2011; Hibbs and Olsson, 2004).

[6]An influential appropriability theory goes back to Carneiro (1970)'s circumscription hypothesis, which contends that states tend to emerge in regions of high agricultural productivity that are geographically enclosed—such that farmers cannot easily escape the demands of an expropriating elite. Empirical support for this mechanism has been provided by Mayoral and Olsson (2025) and Schönholzer and François (2023), who show that geographic circumscription is a strong predictor of early state formation. A related framework is developed by Scott (2009), who distinguishes between agrarian and maritime states. Agrarian states rely on the ability to appropriate staple crops through control over land and labor, while maritime states derive power by taxing or otherwise extracting value from long-distance trade. In this context, Scott highlights strategic chokepoints in trade networks—such as river crossings and narrow sea straits—as "zones of high stateness", where taxation, tolls, and confiscation of trade goods become feasible and profitable. The origins of agrarian states, as hypothesized by Scott, have recently received empirical support from Mayshar, Moav and Pascali (2022), who show that regions where geographical constraints favored the cultivation of cereals—crops that are relatively easy to tax or confiscate—were more likely to transition from tribal societies to chiefdoms and states. Johnson and Earle (2000) similarly distinguish between two paths to complex societies: core agrarian states and peripheral "merchant states". While the former extract surpluses from farmers through the control of productive technologies such as irrigation, the latter rely more on the regulation of markets and the strategic management of trade networks.

[7]Prominent examples of functionalist theories argue that the demand for irrigation infrastructure (see Wittfogel, 1957, Bentzen, Kaarsen and Wingender, 2017, and Allen, Bertazzini and Heldring, 2023), storage systems (Testart et al., 1982), or organized security (Dal Bó, Hernández-Lagos and Mazzuca, 2022) played a key role in state formation.

[8]In the 13th century BC, ironworking began spreading from present-day Bulgaria and Romania, while all major civilizations in Europe, West Asia, and Africa collapsed. Within five decades, almost all cities were destroyed, trade relations were severed, writing systems vanished, and populations living along the coast moved to the interior. Palmer (1962) suggested that the spread of ironworking might be the reason behind the Bronze Age collapse. Iron is superior to bronze for weapons and agricultural tools and can be found almost anywhere in the Old World. The diffusion of the ironworking technology might have been detrimental to the Bronze Age main civilizations for two reasons. First, it undermined the role of Bronze Age cities as it nullified their locational advantage on the trade routes towards copper and tin mines and made obsolete a series of city-institutions constructed to protect interregional trade. Second, it undermined an elite based on the control of metal trade, and it allowed newly formed peripheral armies with iron weapons to quickly destabilize entire states.

The paper is organized as follows. Section 2 describes the data. Section 3 lays out the empirical strategy to identify the passage routes in the metal trade network. Section 4 studies the relation between these passage routes and the rise and diffusion of cities and states. Section 5 describes and tests the appropriability mechanism. Section 6 presents three case studies that are consistent with a metal trade theory of the Urban Revolution. Some concluding remarks close the paper.

2 Data

This section provides a brief overview of the main datasets used in the article. Summary statistics are reported in Online Appendix Table C.1.

2.1 The Urban Revolution

The Urban Revolution was a multifaceted process characterized by two defining elements: the emergence of centralized state societies and the rise of urban settlements. Gordon Childe (1950) proposed ten specific archaeological markers to distinguish, in archaeological sites, the first instances of the Urban Revolution from earlier Neolithic villages: 1) greater population density than any previous settlement, 2) a class of full-time specialists (craftsmen, merchants, officials, priests, etc.) who do not directly procure their own food and live on-site, 3) a system of transfers (through taxes or donations) of the primary surplus from producers to an elite, 4) monumental buildings symbolizing concentrated social surplus, 5) a ruling class exempt from manual tasks, 6) writing, 7) the elaboration of exact, predictive science, 8) sophisticated artistic expression, 9) regular long-distance trade relations, 10) a state organization based on residence rather than kinship. Although most of these traits remain widely accepted as standard archaeological markers of the Urban Revolution, no single dataset coherently records the full range of relevant traits across archaeological sites in the Old World.[9] We therefore rely on multiple, complementary datasets, each capturing different subsets of these features, to approximate the broader pattern of early urban and state formation.

A more detailed description of these data—including summary statistics, spatial coverage maps, and specific data sources—is available in Online Appendix A.1.1.

2.1.1 Urban settlements

We use several datasets to capture the first of Childe's criteria identifying the Urban Revolution: the marked increase in settlement size.

Our benchmark source, compiled by Green et al. (2024), geolocates ancient cities and citadels. Its main advantage over earlier datasets is that it incorporates the latest archaeological discoveries and was assembled by professional archaeologists. The authors do not impose a precise hectare or population threshold; a settlement is classified as a city when it is markedly larger than contemporaneous villages and typically has a population in the order of tens of thousands. Citadels, by contrast, are defined as "fortified palaces without a substantial surrounding urban settlement", and their populations rarely exceed 10,000. Each site is assigned to one of three technological–chronological periods: Neolithic, Bronze Age, and Iron Age.[10] The location of Bronze-Age cities and citadels is depicted

[9] Smith (2009) reassesses Childe's list. According to him, criteria 1, 2, 3, 5, and 10 still describe social processes and institutions that are widely recognized as key developments in the rise of early complex societies. Criteria 4 and 8 require nuance because monuments and sophisticated art appear among Neolithic groups but seldom served as overt symbols of elite power. Criterion 6 applies only to the Old World. While practical sciences and sporadic long-distance exchanges predate cities, major advances in exact sciences and sustained interregional trade with a distinct merchant class arose only with urbanism (criteria 7 and 9).

[10] We include the Chalcolithic (Copper Age) within the Bronze Age for two reasons. First, global chronologies differ, and in

in Figure 1. Based on this dataset, we construct a binary indicator for 1x1-degree grid cells in the Old World with at least one attested Bronze Age city or citadel. According to this measure, only 0.5 percent of all grid cells contain such a site.

The second data source is Reba, Reitsma and Seto (2016), which is assembled from two older sources: Chandler (1987) and Modelski (2003). It is a more standard source in the comparative development literature but omits two decades of new archaeological discoveries. Cities are defined as concentrations of at least 10,000 inhabitants.[11]

Robustness checks in the Online Appendix draw on two additional sources. The first is HYDE 3.1 (Klein Goldewijk et al., 2011), which provides urban-population estimates at millennial intervals after 10000 BC, at high resolution. The second is Lawrence, Hinojosa-Balino and Chelazzi (2025), who systematically map all excavated settlements in Southwest Asia and Egypt that exceeded 10 hectares between 8000 BC and AD 1000.

2.1.2 Archaeological sites

To dig deeper into the archaeological markers of the Urban Revolution, we rely on two archaeological data sources.

The first source is Bahn (2000)'s Atlas of World Archaeology, a collection of detailed maps and descriptions of the world's most relevant archaeological sites. The Atlas covers 7,708 sites spanning from the Paleolithic to Classical antiquity. We use the geocoded version of the atlas provided by Schönholzer and François (2023) and extract the sites that fall into the Paleolithic/Mesolithic, Neolithic, and Bronze Age periods. The result is a list of 559 sites from the Paleolithic/Mesolithic, 585 sites from the Neolithic, and 1,512 from the Bronze Age. For the empirical analysis, we construct a panel dataset indicating the presence of archaeological sites, settlements, burials, and elite burials within each 1×1-degree grid cell in the Old World across the Paleolithic/Mesolithic, Neolithic, and Bronze Age periods.

A drawback of the Atlas is that it is not an academic publication and does not offer a systematic inventory of known sites worldwide. Moreover, it does not reflect discoveries made over the past two decades. For these reasons, we supplement it with a second source: the Pleiades Project, an open-source gazetteer of ancient sites (Bagnall, 2022). We restrict attention to sites with precisely known locations and classify them into four categories: settlements, non-elite burials, elite burials, and other sites. To assign these sites to time periods, we use two complementary approaches. First, we rely on the textual descriptions in the Pleiades database to distinguish between pre-Bronze and Bronze Age sites, resulting in a two-period grid-cell panel dataset with 866 sites, 213 of which predate the invention of bronze. Second, we use the numeric start and end dates provided in the database to construct a panel covering the years 7000–1000 BC at 1,000-year intervals. For each millennium, we count how many settlements and archaeological sites are located in each grid cell, yielding a six-period grid-cell level panel dataset.

several national traditions—most notably in Britain and Ireland, where the Beaker-phase Copper Age is routinely treated as the opening stage of the Bronze Age—the Chalcolithic is already subsumed under the Bronze Age. Second, from a theoretical standpoint, the Copper Age already displays unprecedented long-distance trade and complex hierarchies, making it analytically continuous with the Bronze Age.

[11] In our analysis, we rely only on the location of cities, not their estimated populations. Population figures are highly debated in the archaeological literature; we therefore use only the binary information on whether a settlement with 10,000+ inhabitants is attested at any time prior to 1300 BC, the conventional endpoint of the Bronze Age in our study.

2.1.3 China

Additional data on the rise and spread of the Urban Revolution in China are drawn from the Atlas of Chinese Cultural Relics, the most comprehensive catalog of Chinese archaeological sites. This source is based on joint efforts between the Chinese central and provincial cultural heritage bureaus to compile a mapping of all excavated monuments and remains in China. The catalog is complete for 28 out of 34 provinces (data for the provinces of Taiwan, Macau, Hong Kong, Jiangxi, Guizhou, and Hainan are still under construction). Each archaeological site is geocoded, dated, and described in detail. We OCR the volumes that comprise the Atlas and focus on the 85,000+ sites dating before the Common Era. Each site contains a long description (on average 58 words). For approximately a tenth of the sample, we manually coded whether the description references any of the following: 1) large settlements, 2) monumental buildings, 3) ruling class exempt from manual tasks, 4) a standardized system of measures and records used in exchange and production, 5) writing, 6) highly developed art forms. After developing this training sample, we estimate six separate naïve Bayesian classifiers, one for each criterion, and use them to classify all sites in the Atlas. This procedure allows each site to satisfy multiple criteria. Once each site has been classified, we then count the sites associated with each criterion in each 1×1 grid cell dating from the Paleolithic, Neolithic, and Bronze Age.

2.1.4 Europe

During the Bronze Age, Europe saw the rise of chiefdoms marked by unprecedented social inequality. To capture this transformation, we compile data on elite burials containing metal weapons—an established archaeological marker of local chiefdoms (Earle et al., 2015; Renfrew and Bahn, 2016, pp. 175). Our main source is the Prähistorische Bronzefunde (PBF), a systematic catalog of European bronze artifacts compiled since 1966. We supplement it with the Bronze und Urnenfelderzeitliche Schwerter database by Hahnekamp (2011). Together, these sources offer broad coverage of Central and Western Europe.

For each artifact, we record three key details: (i) the geographic coordinates of the excavation site; (ii) the context of the site (e.g., whether it is a burial); and (iii) the artifact's archaeological type or family (e.g., Pădureni axe type). From (ii), we identify elite burials; from (iii), we assign the burial to a specific phase of the Central European Bronze Age (Early, Middle, or Late).

We then construct a panel dataset on a 1×1-degree grid, distinguishing two time periods: the Early/Middle Bronze Age [3000–1300 BC) and the Late Bronze Age [1300–750 BC]. For each grid cell and time period, we count the number of elite burials.

2.2 Bilateral trade flows

We reconstruct prehistoric trade flows using archaeological artifacts. To be included in our analysis, artifacts must satisfy two criteria. First, both the find site and the provenance of the artifact must be identifiable, allowing us to infer trade flows by assigning the excavation site as the destination and the production site as the origin. Second, the period of production must be attributable to either the pre-Bronze Age or the Bronze Age. We identify approximately 7,500 artifacts that satisfy these criteria, distributed roughly evenly across the pre-Bronze Age and Bronze Age periods. The data are drawn from Flückiger et al. (2022), who compile a dataset covering both periods using established databases (Pétrequin et al. (2012); Schauer et al. (2020)), as well as a range of additional academic sources. A more detailed description of the data, including the full list of primary sources

and maps illustrating spatial coverage, is provided in Online Appendix A.1.2. Using geocoded information on artifact provenance and excavation sites, we assign each item to its corresponding 1×1-degree grid cell of origin and destination. We then aggregate these data at the grid-cell-pair level, generating, for each time period, a count of distinct find sites in grid cell j that show evidence of goods produced in grid cell i. We use this site-based measure, rather than artifact counts, because artifact counts may be distorted by hoards and differences in excavation intensity across sites. All grid-cell pairs fall within the geographical scope of Europe and the Mediterranean Basin.

2.3 Cropland and productive land

Data on the estimated area of cropland (in km^2) within each 1×1-degree grid cell in 3000 BC is provided by Klein Goldewijk et al. (2011).

The distribution of farming populations may have been shaped by the influence of early civilizations. To isolate exogenous variation in this variable, we use a specific measure of land productivity, the net primary production (NPP). This measure gauges the potential biomass that can be produced in a region based on local geo-climatic conditions, and is therefore unaffected by human intervention. We utilize the Miami model (Lieth, 1975) to determine the NPP for given local climatic conditions, and the CHELSA-TraCE21k simulator (Karger et al., 2021) to reconstruct climatic conditions at high spatial resolution between 4000 BC and 3000 BC.[12] This measure of land productivity has several advantages. First, it captures land productivity for both hunter-gatherers and farmers. Second, it is independent of the mix of crops and animals used for subsistence, whether domesticated or not. Additional details are provided in Online Appendix A.1.3.

2.4 Metal mines and deposits

We conducted a comprehensive review of the archaeological literature to compile a new dataset on Bronze Age metal mining sites. A site is included in the dataset only if both its location and its activity during the Bronze Age can be clearly identified. The final dataset contains 46 tin mines and 121 copper mines, distributed across 32 and 80 separate grid cells, respectively. For mines supplying Europe, we further distinguish between those operating before and after 1300 BC. Additional details on data construction, data sources, and a map showing the location of these mines can be found in Online Appendix A.1.4.

Using these data in empirical analysis presents two challenges. First, the location of mines may be endogenous to the spatial distribution of cities and states. Second, the archaeological records of Bronze Age mining activity do not accurately reflect the full set of exploited metal deposits. This is particularly true for tin: the sources used during the third millennium BC remain uncertain, as Bronze Age miners typically relied on alluvial cassiterite, which leaves little or no geological trace. Measurement error is thus non-random, as regions with more intensive archaeological excavation are more likely to yield evidence of ancient mining. To address these concerns, we also use geological data on the location of tin and copper deposits as they are known today, regardless of whether they have ever been exploited or not. These data are sourced from the U.S. Geological Survey (2005).

[12] More precisely, we use the equations from the original Miami model (Lieth, 1975), applying the updated coefficients from Zaks et al. (2007). See Appendix A.1 for more details.

2.5 Additional data

We rely on several additional datasets, which are described in detail in Online Appendix A.1.5. First, we construct a Bronze Age transport cost surface by aggregating data on landmasses, inland waterways, and mountainous terrain at the 0.25×0.25-degree grid cell level. Second, to capture organized warfare, we compile a dataset on prehistoric weapons—dating to the Bronze Age or earlier—based on the catalogs of major museum collections. Third, we collect a range of geographic and environmental variables at the grid-cell level, including temperature, precipitation, growing season length, malaria suitability, biome classification, and proximity to natural harbors.

3 Bronze Age transit regions

This section aims to identify the key transit regions within the Bronze Age metal trade network. In the first subsection, we estimate the mode-specific relative transport costs using established methods from the international trade literature, applied to Bronze Age exchange data. In the second, we combine these estimates with information on the spatial distribution of cropland and metal sources to locate the road-knots in the metal trade network. Because the placement of Bronze Age cropland and mining sites was likely shaped by the spatial diffusion of early state societies, we isolate an exogenous element of the road-knots in the third subsection. Specifically, we identify the transit regions that connect metal deposits with naturally productive land—two features determined by environmental conditions rather than human intervention.

3.1 Period-specific transport costs

To our knowledge, estimates of relative transportation costs predating the Iron Age are unavailable. We infer these costs using a methodology in the spirit of Donaldson (2018). Intuitively, our objective is to identify the combination of mode-specific transport costs that best fits the observed trade patterns. To this end, we begin by dividing the world into transport-surface grid cells of 0.25×0.25 degrees and assigning each cell to one of three transport modes: sea, river, or land.[13] Each mode is associated with a per-unit-distance transport cost, $\boldsymbol{\alpha} = \left(\alpha^{sea}, \alpha^{river}, \alpha^{land}\right)$. We impose two restrictions that reflect the technological constraints of the period. First, maritime transport is permitted only along the coast (and below 60° latitude). Second, transport is assumed to be infeasible across high mountain ranges (above 15,000 feet).

The vector $\boldsymbol{\alpha}$ is unknown and is therefore estimated from the data. Conditional on $\boldsymbol{\alpha}$, we use Dijkstra's algorithm (Dijkstra, 1959) to compute the least-cost path between every origin–destination grid-cell pair od, and let $LC_{od}[\boldsymbol{\alpha}]$ denote the corresponding transportation cost.[14] We then estimate a standard gravity equation using the Poisson pseudo-maximum likelihood (PPML) estimator:

$$\mathcal{X}_{od} = \exp\left(\delta \ln LC_{od}[\boldsymbol{\alpha}] + \beta_o + \beta_d\right) + \varepsilon_{od}, \tag{1}$$

where $\mathcal{X}_{od}$ denotes the number of period-specific find sites in grid cell d that show evidence of trade originating from cell o. β_o and β_d are a full set of origin and destination fixed effects, and ε_{od} is an additive error term.

We estimate equation (1) iteratively over all relative transport cost combinations $\alpha^{sea} = [1, 20]$, $\alpha^{river} = [1, 50]$, and $\alpha^{land} = [1, 100]$ for two sets of artifacts: those dating from the Stone Age (SA)

[13] The relatively coarse cell size is chosen to account for the fact that coastlines and river courses may shift over time.

[14] We assume that transshipment across transport modes is costless.

and those dating from the Bronze Age (BA).[15] For each set of artifacts, we then choose the vector $\boldsymbol{\alpha}$ that maximizes the log-likelihood of the estimated gravity equation. We normalize $\alpha^{sea} = 1$, so that $\boldsymbol{\alpha}$ captures how costly a given shipping mode is relative to maritime shipping. For the Bronze Age, this fit-maximizing transport cost vector is given by $\widehat{\boldsymbol{\alpha}_{\boldsymbol{BA}}} = (1, 2, 6)$, while for the Stone Age it is $\widehat{\boldsymbol{\alpha}_{\boldsymbol{SA}}} = (1, 1/14, 2/7)$.

These results suggest that the relative costs of shipping goods changed substantially across modes after the invention of bronze. Sea shipping shifted from being the most to the least costly mode. This aligns well with archaeological evidence. Before the Bronze Age, waterborne transport relied on dugout canoes—shallow-hulled and oar-driven—unsuitable for long-distance or large-scale transport. Consequently, long-distance trade occurred mostly overland. For instance, exchanges between the Near East and Mediterranean Europe likely relied on the Anatolian land bridge (Rahmstorf, 2010). This pattern shifted with the advent of deep-hulled plank boats and sail technology in the Early Bronze Age. These vessels could carry several tonnes of goods and were stable enough for coastal shipping. Sailing innovations further reduced costs by enhancing maneuverability and speed. These changes enabled large-scale maritime trade, making sea routes the most cost-effective for long-distance transport (e.g., Nessel, Neumann and Bartelheim, 2018). This fostered economic integration between the Mediterranean and Mesopotamia and facilitated trade between Mesopotamia and the Indus civilization via the Indian Ocean (e.g., Rahmstorf, 2010; Cunliffe, 2011, p. 189; Vogt, 1996). Our estimates also suggest a moderate decline in overland transport costs relative to river transport between the Stone and Bronze Ages. While river transport remained the cheaper option in both periods, the cost gap between river and land transport narrowed over time. This, too, aligns with archaeological findings. In the Stone Age, goods were transported overland by humans or ox carts, resulting in slow and low-capacity movement. During the Bronze Age, several innovations improved overland speed and freight capacity—such as the invention of the spoked wheel and chariot technology (Uckelmann, 2013). In arid regions, the domestication of camels further enabled the bulk transport of goods.

3.2 Identifying the road-knots: the construction of the metal transit index

We adopt a three-step approach to identify road-knots in the metal trade transportation network.

First, using the estimated Bronze Age transport costs ($\widehat{\boldsymbol{\alpha}}$) and the topographic map, we compute the least-cost paths connecting each grid cell with cropland (in 3000 BC) to each Bronze Age tin and copper mine. To exclude impractically long routes, we discard paths for which total transport cost exceeds the cost of traveling a distance equivalent to 10,000 kilometers by sea.[16]

Second, for each cropland cell f, we select the tin mine and the copper mine with the lowest transport cost to f. Denote these selected mines by $Sn^{\star}(f)$ and $Cu^{\star}(f)$, respectively, and let $P^{\star}_{Sn^{\star}(f),\,f}$ and $P^{\star}_{Cu^{\star}(f),\,f}$ be the corresponding least-cost paths from these mines to cell f.

Third, for each grid cell k, we compute a "metal transit index", T_k, as the cropland-weighted count of these selected least-cost paths that traverse k:

$$T_k = \sum_{f \in \mathcal{F} \setminus \{k\}} \left(\mathcal{I}_{P^{\star}_{Sn^{\star}(f),\,f}}[k] + \mathcal{I}_{P^{\star}_{Cu^{\star}(f),\,f}}[k] \right) \text{Crop}_f \,. \tag{2}$$

[15] Throughout, we loosely use the term "Stone Age" to describe the prehistoric period before the advent of bronze technology.
[16] This threshold has negligible impact on the results.

where $\mathcal{F}$ is the set of all grid cells with cropland; $\mathcal{I}_{P^{\star}_{Sn^{\star}(f),f}}[k]$ is an indicator function equal to 1 if $k \in P^{\star}_{Sn^{\star}(f),f}$; $\mathcal{I}_{P^{\star}_{Cu^{\star}(f),f}}[k]$ is an indicator function equal to 1 if $k \in P^{\star}_{Cu^{\star}(f),f}$; $Crop_f$ is the cropland area (measured in square kilometers) in grid cell f in 3000 BC. T_k thus measures the square kilometers of cropland for which the cost-minimizing routes to access tin and copper mines traverse cell k. It is conceptually similar to betweenness centrality in network analysis.[17]

The distribution of T_k is highly skewed: for 69 percent of grid cells, T_k equals zero. Figure 1 displays the spatial distribution of T_k, with darker areas indicating higher values. We observe a clear correlation: most Bronze Age cities (black dots) tend to cluster in metal trade road-knots.

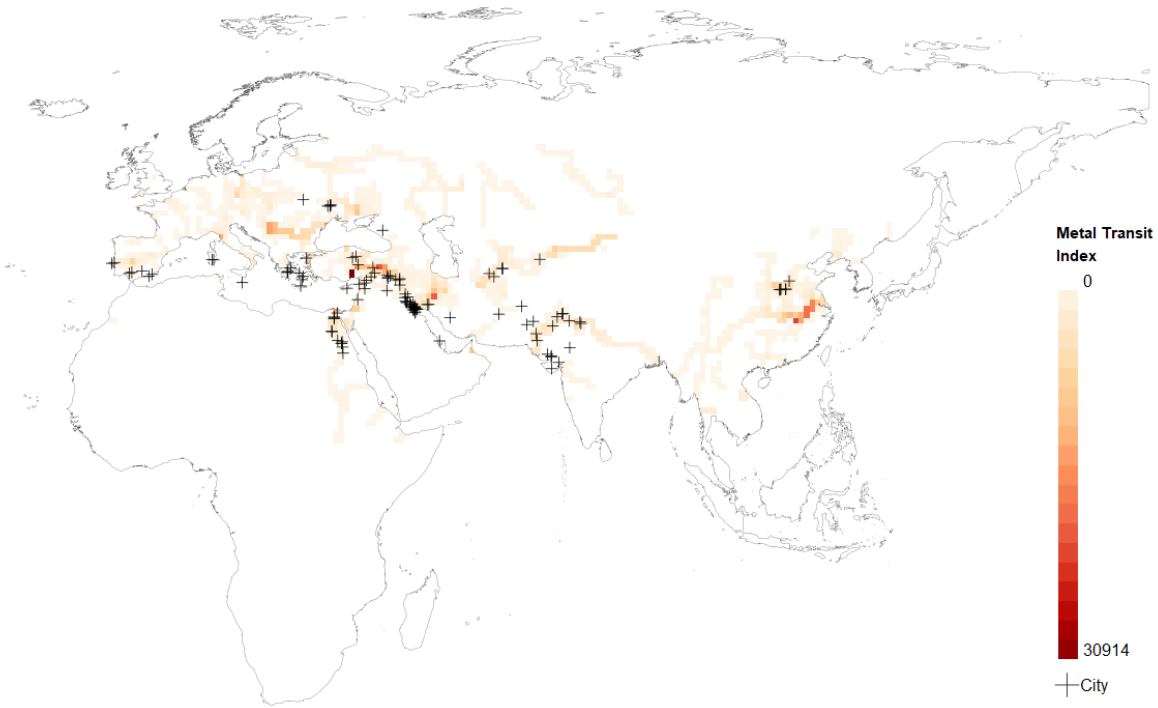


Figure 1: Geographic scope of analysis, Bronze Age cities and metal transit index.
Figure depicts the location of Bronze Age cities and citadels (sources: Green et al. (2024) and Reba, Reitsma and Seto (2016)) along with the metal transit index (as defined by equation (2)).

3.3 Isolating an exogenous component: the construction of the natural metal transit index

The association observed above does not necessarily imply causality: the distribution of Bronze Age mines and cropland—used to construct T_k—may have been shaped by the spatial diffusion of cities. To isolate an exogenous variation in the metal transit index, we construct an alternative centrality measure, D_k, which captures transit regions connecting metal deposits (rather than Bronze Age mines) with productive land, measured by net primary production (rather than Bronze Age cropland)—two components determined by environmental conditions rather than human intervention. We hereafter refer to this measure as the "natural metal transit index". Formally, we define

[17] The betweenness centrality of a node in a network is defined as the number of least-cost paths that pass through that node connecting all node-pairs within the network. See Appendix D.8.1 for details.

it as:

$$D_k = \sum_{f \in \mathcal{F}} \left(\mathcal{I}_{P^{\star}_{DSn^{\star}(f),\,f}}[k] + \mathcal{I}_{P^{\star}_{DCu^{\star}(f),\,f}}[k] \right) \text{Prod}_f \,. \tag{3}$$

where $DSn^{\star}(f)$ and $DCu^{\star}(f)$ denotes the tin and copper deposits with the lowest transport cost to f; and $Prod_f$ is the net primary production of cell f in 3000 BC.

Unlike the metal transit index T_k, the natural metal transit index D_k is strictly positive for all grid cells. This reflects the fact that even the most ecologically marginal environments can sustain a minimal level of biomass, thereby enabling some degree of connectivity.

It is important to note that the estimation of D_k relies on the relative transport costs estimated for the Bronze Age. The available technology during that time might have also been influenced by the Urban Revolution itself. To account for this, we also calculate an alternative value for D_k, which is based on the relative transport costs estimated for the Stone Age.

In Online Appendix B, we describe a wide range of additional centrality measures in the natural transportation network, which are used as control variables in the empirical analysis. In particular, we explain the construction of the proximity-to-cropland and proximity-to-mines variables.[18]

4 Metal trade and Urban Revolution

4.1 Cross-sectional evidence: Bronze Age cities

This subsection examines the link between metal trade and Bronze Age urban settlements. As we have seen, a visual inspection of Figure 1 provides a first clue about the positive correlation between the location of Bronze Age road-knots and cities. We formally test for this relationship by estimating the following equation:

$$y_k = \theta \; IHS(T_k) + X_k' \, \phi + \varepsilon_k, \tag{4}$$

where y_k is a dummy indicating whether a city had emerged in grid cell k during the Bronze Age.[19] The main regressor of interest is T_k, the metal transit trade index of grid cell k, transformed using the inverse hyperbolic sine (IHS) transformation. The IHS transformation behaves similarly to a logarithm—filtering out scale effects and reducing the influence of outliers—while allowing us to retain zero-value observations. As the IHS transformation complicates the interpretation of coefficient magnitudes, we also present results using the untransformed variable. X_k is a vector of controls.

Appendix Figure C.1 displays a binned scatterplot showing the relationship between the dependent variable and our main regressor of interest. The association appears clearly positive. In Table 1, we assess this relationship more formally. Column (1) reports the baseline OLS estimates, controlling only for continent fixed effects. In column (2), we add a set of controls that account for geographical features that are used in the construction of the metal transit index: proximity to mines, proximity

[18] Formally, proximity to cropland of grid cell k is defined as: $PC_k = \sum_{f \in \mathcal{F} \setminus \{k\}} \frac{1}{LC_{f,k}} \times Crop_f$ where $LC_{f,k}$ denotes the shipping costs from grid cell f to cell k and $Crop_f$ is the croplands area of cell f. Proximity to mines is instead defined as: $PM_k = 1/\left(LC_{Sn^{\star}(k),k} + LC_{Cu^{\star}(k),k}\right)$, where $LC_{Sn^{\star}(k),k}$ and $LC_{Cu^{\star}(k),k}$ are the costs associated with shipping tin and copper from the cost-minimising mines producing these metals to grid k, respectively.

[19] For ease of interpretation, we multiply the dummy by 100.

Table 1: Metal transit trade and Bronze Age cities

	OLS		Probit		2SLS IV			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Panel A: Second stage Green et al. (2024): Any city or citadel (×100)							
Metal transit index (IHS)	0.344	0.353	0.156	0.086	0.407	0.483		
	(0.074)	(0.076)	(0.025)	(0.028)	(0.089)	(0.126)		
	[0.094]	[0.097]	[0.009]	[0.012]	[0.123]	[0.164]		
Metal transit index							0.002	0.003
							(0.001)	(0.001)
							[0.001]	[0.001]
	Panel B: Second stage Reba, Reitsma and Seto (2016): Any city (×100)							
Metal transit index (IHS)	0.383	0.424	0.201	0.092	0.342	0.312		
	(0.087)	(0.093)	(0.029)	(0.033)	(0.083)	(0.095)		
	[0.134]	[0.141]	[0.011]	[0.012]	[0.124]	[0.127]		
Metal transit index							0.002	0.002
							(0.000)	(0.001)
							[0.001]	[0.001]
					Panel C: First stage			
Natural metal transit index (IHS)					0.490	0.308	86.183	43.091
					(0.031)	(0.030)	(13.032)	(8.275)
					[0.051]	[0.048]	[18.407]	[11.294]
Geography controls	No	Yes	No	Yes	No	Yes	No	Yes
Continent dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	10,970	10,970	10,970	10,970	10,970	10,970	10,970	10,970
Mean dependent variable Panel A	0.483	0.483	0.483	0.483	0.483	0.483	0.483	0.483
Mean dependent variable Panel B	0.419	0.419	0.419	0.419	0.419	0.419	0.419	0.419
R-squared Panel A	0.016	0.018	0.177	0.259				
R-squared Panel B	0.022	0.028	0.254	0.422				
First-stage F-stat (5×5 grids)					251.000	102.261	43.734	27.115
First-stage F-stat (Conley 1000km)					92.374	40.999	21.690	14.389

Notes: Panels A and B report cross-sectional OLS, probit, and 2SLS IV estimates of equation (4). Panel C reports the corresponding first-stage estimates (equation (5)). The unit of observation is a 1×1-degree grid cell k. 'Any city or citadel' equals 100 if a city or citadel was present in grid cell k during the Bronze Age (and 0 otherwise). 'Any city by 1300 BC' equals 100 if a city was present in grid cell k by 1300 BC (and 0 otherwise). 'Metal transit index' is defined by equation (2), while 'Natural metal transit index' is defined by equation (3). Footnotes 18 and 21 describe the construction of the controls: Geography controls include proximity to mines, proximity to croplands, sea dummy, river dummy, mountains dummy, and centrality in the transport network. Standard errors clustered at the level of 5×5 degree grid cells are reported in parentheses. Standard errors computed using the approach of Conley (1999) (cut-off 1000km) are reported in square brackets.

to cropland, topography[20], and centrality in the natural transportation network.[21] The coefficient on the metal transit index is estimated to be positive and statistically significant, with only minimal change after including controls. Columns (3) and (4) report results from probit regressions, which better account for the binary nature of the dependent variable and the rarity of Bronze Age cities in the cross-section. The estimated marginal effects are similar to those from the OLS regressions when the metal transit index is zero (its median value). Notably, the probit model fits the data substantially better: the R^2 is more than ten times larger than that in the corresponding OLS specification.

Both OLS and probit estimates are far from having a causal interpretation. The metal transit index is influenced by various factors, including the spatial distribution of croplands and mines, which themselves may be affected by the spatial distribution of Bronze Age cities and civilizations. To get closer to a causal relationship between the metal transit trade and the emergence of cities, we employ a 2SLS estimation framework. The first stage is the following:

$$IHS(T_k) = \delta \, IHS(D_k) + X_k' \, \eta + \zeta_k, \tag{5}$$

where D_k denotes the natural metal transit index.

Appendix Table D.3 presents the reduced-form estimates, while Appendix Figure C.1 shows the corresponding binned scatterplot. The IV estimates are reported in columns (5) and (6). The first-stage results, shown in Panel C, indicate that the instrument is strongly and positively correlated with the metal transit index, with an F-statistic well above conventional thresholds. The second-stage estimated coefficient θ is positive and statistically significant. It is about 25 percent larger than the corresponding OLS estimates. This difference suggests that the OLS estimates may be biased not only by endogeneity but also by classical measurement error. Columns (7) and (8) re-estimate the model using the metal transit index in levels rather than its IHS transformation. The coefficient on T_k remains positive and statistically significant. An increase of 170 square kilometers in cropland connected to a mine via a route traversing the grid cell–corresponding to an increase from 0 to the mean of the metal transit index–raises the probability that a Bronze Age city appears in the cell by 0.34 percentage points in the specification without additional controls and by 0.51 percentage points in the specification with controls. These are substantial effects, given that Bronze Age cities appear in only 0.5 percent of grid cells in the sample.

Additional analyses and robustness checks

We conduct a series of additional analyses, which are described in detail in the Online Appendix. Below, we provide a brief overview.

1. The transition from arsenic bronze to tin bronze (Online Appendix D.1.3)

Reba, Reitsma and Seto (2016) provide city-level data at century intervals. Using this dataset, we re-estimate our 2SLS regressions with the dependent variable capturing the presence of cities over successive 500-year intervals from 3300 BC to 1300 BC. This approach allows us to trace the relationship between metal transit trade and city formation over time. The results suggest that the estimated effect of metal trade routes on urban emergence is already positive by 3300 BC,

[20] We include separate dummy variables for each topographical characteristic that influences the estimated transport cost of traversing a cell. More specifically, we include indicators for coastal cells, riverine cells, and cells with an average elevation above 15,000 feet

[21] Network centrality is measured as the eigenvector centrality within the natural transportation network.

although it becomes statistically significant only from 2800 BC onward. Moreover, we distinguish between transit routes to copper and tin mines. Copper-related routes appear important from the fourth millennium onward, while tin-related routes gain relevance only at the very end of the third millennium. This likely reflects the technological transition from arsenic bronze to tin bronze during that period.

2. The endogeneity of relative transportation costs (Online Appendix D.1.4)

Our identification strategy relies on relative transportation costs estimated from Bronze Age trade patterns in Europe and the Middle East. One concern is that such costs may reflect transport technologies tailored to the needs of already established civilizations, rather than exogenous geographic constraints. To address this, we implement two checks. First, we restrict our sample to regions not used in estimating the transportation cost vector and find that our results remain stable. Second, we reconstruct the instrument using relative transportation costs estimated for the Stone Age. The results are qualitatively unchanged, although—as expected—in the second exercise the instrument exhibits somewhat reduced statistical power.

3. Exploiting data on megasites (Online Appendix D.1.5)

The analysis presented in this section showed that Bronze Age urban settlements tend to emerge along metal transit corridors. One concern, however, is that this pattern may simply reflect continuity from earlier Neolithic settlement patterns rather than a structural shift linked to the emergence of metallurgy.

To evaluate this, we first exploit the fact that Green et al. (2024) provide data not only on large settlements during the Bronze Age but also on large Neolithic settlements—commonly referred to as "megasites" in the archaeological literature. Using 2SLS cross-sectional regressions, we find no significant correlation between the metal transit index and the presence of large settlements during the Neolithic. In contrast, using a 2SLS difference-in-differences design, we find that large settlements became significantly more likely to emerge along metal transit corridors during the transition to the Bronze Age. A key limitation of this exercise is the relatively small number of large Neolithic settlements in the dataset (only 16 observations in the pre-period).

As a further robustness check, we repeat the analysis using data from Lawrence, Hinojosa-Balino and Chelazzi (2025), which focus on the size of settlements in Southwest Asia and Egypt. This dataset provides a continuous measure of settlement area, allowing us to examine the intensive margin of urban development. However, its geographic scope is more limited. The results align with our earlier findings: in cross-sectional 2SLS regressions, the metal transit index does not predict the size of pre-Bronze Age settlements, while 2SLS difference-in-differences estimates show a marked increase in settlement size along metal corridors during the Bronze Age.

Together, these two exercises suggest that metal trade was not simply layered onto pre-existing Neolithic geographies. Instead, it seemed to have contributed to a substantive reshaping of the urban landscape in both location and scale.

4. Digging deeper into the metal transit index measure (Online Appendix D.1.6)

Finally, we assess whether our findings are driven by the specific way we construct the metal transit index. We compare our benchmark measure—which counts least-cost paths linking cropland to copper and tin mines—with several alternative versions, including indices that (i) capture general connectivity to farmland or to mines, and (ii) rely on shortest rather than least-cost paths. While

these alternative measures are moderately associated with city presence, the results show that our metal transit index remains the strongest and most robust predictor of urban development. This suggests that it was not general connectedness, but rather the specific role of a location as a metal-agriculture transit hub, that mattered most.

5. Other traded goods and pre-Bronze Age trade (Online Appendix D.4)

Copper and tin were not the only goods traded during the Bronze Age, but comparable spatial data are unavailable for other commodities such as textiles, timber, and spices. To test whether our results reflect a general effect of transit trade rather than a mechanism specific to metals, we construct a generic natural transit index. This measure captures the number of least-cost paths that pass through each cell when connecting all possible pairs of grid cells with productive land across the Old World—independent of the distribution of cropland and metal resources. The estimated coefficient on the metal transit index remains effectively unchanged when both indices are included in the regression.

We also examine obsidian, a volcanic glass that has received substantial attention in the archaeological literature and was widely traded before and during the Bronze Age. Using the spatial distribution of known geological deposits and productive land, we construct a natural obsidian transit index. As with the generic transit index, controlling for the natural obsidian transit index does not affect the estimated coefficient on the metal transit index.

Taken together, these findings suggest that our results are likely to be driven by copper and tin transiting trade rather than by other traded goods.

6. Land productivity and agriculture (Online Appendix D.1.7)

A longstanding view in the literature emphasizes the role of agriculture in the emergence of cities. To assess this channel, we replicate our 2SLS analysis controlling for several measures of land productivity commonly used in the comparative development literature. The inclusion of these controls leaves our main finding unchanged in qualitative terms: the coefficient on the metal transit index remains positive, statistically significant, and economically meaningful across specifications. By contrast, the relationship between land productivity and city presence is sensitive to the proxy and sample used. While some land-productivity measures are statistically significant in individual specifications, the estimated coefficients vary in sign and magnitude and do not reveal a stable pattern. At the same time, although Bronze Age cities were often located in areas with limited agricultural productivity, they generally retained access to fertile land nearby. Consistent with this, our measure of proximity to cropland—which is included as a control in Table 1—is positively and statistically significantly associated with city presence.

7. Heterogeneous effects (Online Appendix D.1.8)

To further investigate the interaction between transit trade and access to key resources in shaping the geography of Bronze Age urbanization, we examine whether the association between the metal transit index and the presence of Bronze Age cities varies by proximity to fertile cropland and metal mines. OLS and 2SLS estimates indicate that this relationship is positive and large only in regions located near fertile cropland (i.e., those above the median in the proximity-to-cropland distribution), while it is considerably weaker and statistically insignificant in more remote areas. A similar pattern holds for proximity to metal mines. The 2SLS results should, however, be interpreted with caution, as the corresponding first-stage F-statistics fall below conventional threshold values. With this caveat, the results suggest a complementarity between metal transit trade and access to

both agricultural and metallurgical resources in determining where early cities emerged.

8. Standard errors (Online Appendix D.1.9)

To account for spatial autocorrelation in Table 1, we compute standard errors using two approaches: clustering at the 5×5 grid cell level and applying Conley (1999) standard errors with a 1,000 km distance cutoff. In the Online Appendix, we demonstrate that varying the clustering grid size (from 1×1 to 20×20) or the Conley cutoff radius (from 100 km to 2,000 km) has only a modest impact on the estimated standard errors. In all cases—both for OLS and 2SLS specifications—the coefficient on T_k remains statistically significant.

9. Other robustness checks (Online Appendix D.1.10)

(a) Given the rarity of Bronze Age cities—they appear in only 0.48 percent of grid cells—we verify robustness when restricting our analysis to areas with some urban activity. Results hold when (i) excluding Sub-Saharan Africa and Russia; (ii) retaining only UN subregions with at least one city; (iii) retaining only countries with at least one city; (iv) restricting the spatial extent to the minimum and maximum latitude and longitude of observed cities; or (v) limiting the analysis to the area enclosed by the convex hull of city locations.

(b) To ensure that our results are not driven by omitted environmental factors, we include as additional controls all geographic variables used in Henderson et al. (2018), which specifically investigates the geographic determinants of economic activity over time. Our findings remain robust to the inclusion of these controls.

(c) We test robustness with respect to the grid size by re-running the analysis using 0.5×0.5 degree cells. Results remain consistent.

(d) We re-estimate equation (4) using IV probit. The coefficients remain statistically significant and are larger in magnitude than those obtained with standard probit.

(e) We obtain qualitatively equivalent results when we use urban population data from HYDE 3.1.

Overall, the results presented in this subsection provide evidence supporting the hypothesis that transiting long-distance metal trade played an important role in explaining the onset of the Urban Revolution. Cities were much more likely to be established along trade corridors connecting cropland with tin and copper mines. The 2SLS estimates suggest a causal link, but, due to the cross-sectional nature of the data, we cannot exclude the possibility that omitted geographic factors might be driving the results. To overcome this limitation, we next turn to panel data, which allow us to account for time-invariant geographic factors.

4.2 Difference-in-differences evidence: archaeological sites

In this subsection, we draw on two data sources, Bahn (2000) and Bagnall (2022), which report the location of archaeological sites and settlements across the Old World prior to the Bronze Age Collapse. We use a difference-in-differences design to test whether the spatial distribution of archaeological sites shifted toward metal transit routes during the transition from earlier prehistoric periods to the Bronze Age. Specifically, we estimate the following equation:

$$y_{k,p} = \beta\, IHS(T_k) \times I_p^{BA} + X'_{k,p}\,\Phi + \mu_k + \mu_{c,p} + \psi_{k,p}, \tag{6}$$

where $y_{k,p}$ captures the presence of an archaeological site dating from period p in grid cell k; I_p^{BA} is an indicator for the Bronze Age; T_k is the metal transit index; $X_{k,p}$ is a set of time-varying control

variables. We include cell fixed effects, μ_k, and continent×period fixed effects, $\mu_{c,p}$. To address potential endogeneity, we employ a 2SLS specification, with the first stage given by:

$$IHS(T_k) \times I_p^{BA} = \sigma\, IHS(D_k) \times I_p^{BA} + X'_{k,p}, \psi + \rho_{c,p} + \rho_k + \xi_{k,p}. \tag{7}$$

Columns (1) and (2) of Table 2 report OLS results. Column (1) includes no additional controls, while column (2) incorporates the full set of geographic covariates—proximity to mines, cropland, topography, and network centrality—each interacted with period fixed effects. Columns (3) and (4) show the corresponding 2SLS estimates. In both archaeological datasets, we find that road-knots, as captured by the metal transit index, are associated with a significant increase in the probability of observing archaeological sites during the Bronze Age relative to earlier periods. Columns (5) and (6) present 2SLS estimates using the untransformed transit index. To help interpret these coefficients, note that in areas where the metal transit index equals zero, the number of cells with Bronze Age archaeological sites is roughly the same as the number of Neolithic sites. The coefficient in column (5) then suggests that a substantial share—nearly half—of the increase in the share of grid cells containing archaeological sites from the Neolithic (2.86 percent) to the Bronze Age (4.48 percent) can be attributed to the metal transit index.[22]

Figure 2 presents analogous 2SLS difference-in-differences estimates disaggregated by site type. Results are robust when focusing only on settlements. Interestingly, the metal transit index is not significantly associated with an increased presence of burials with the onset of the Bronze Age, but is positively and significantly associated with an increase in elite burials (e.g., pyramids or tholos tombs), an archaeological marker of early social hierarchy.

Pre-trends

To trace out the impact of metal transit trade over time and test for pre-trends, we construct two panel datasets. The first, based on Bahn (2000), includes three periods: Paleolithic/Mesolithic, Neolithic, and Bronze Age. The second, from Bagnall (2022), has 1,000-year intervals from 7000 to 1300 BC. We interact the metal transit index with period dummies, using the pre-Bronze Age period as the reference category. For the three-period panel, this is the Neolithic; for the higher-frequency panel, it is the interval [5000,4000) BC. Our specification is:

$$y_{k,p} = \sum_{p=1}^{P} \psi_p\, IHS(T_k) \times I_p + X'_{k,p}\, \lambda + \tau_{c,p} + \tau_k + \varepsilon_{k,p}. \tag{8}$$

where $\tau_{c,p}$ are continent-by-period fixed effects and τ_k are cell fixed effects.

Figure 3 displays the resulting 2SLS point estimates. The metal transit index is associated with an increased likelihood of observing archaeological sites in a grid cell only after the onset of the Bronze Age, with no evidence of pre-trends. Panel (b) shows that this association becomes statistically significant starting in the second millennium BC.

Additional analyses and robustness checks

1. Robustness checks

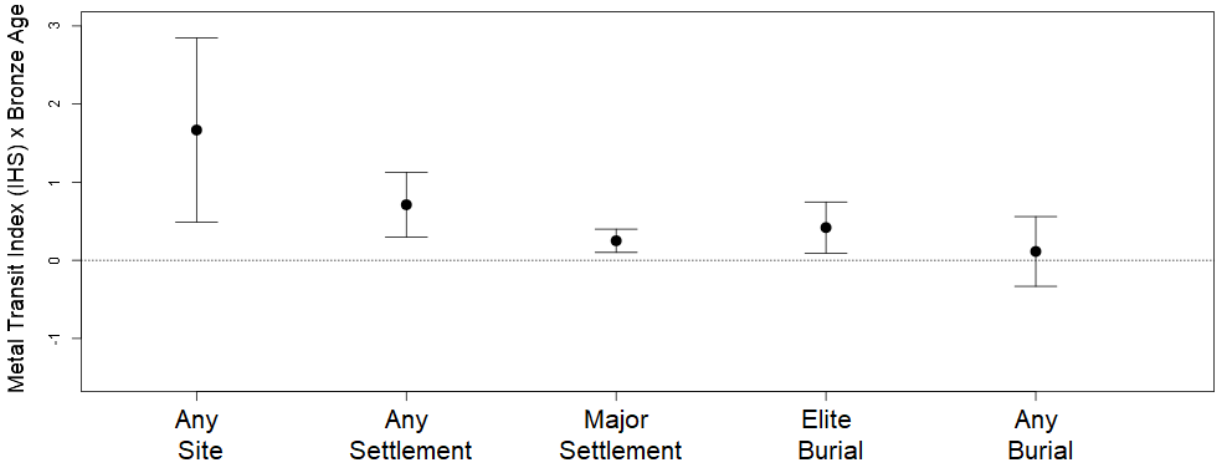
[22]The observed increase between the Neolithic and the Bronze Age is 1.62 percentage points. Evaluated at the sample mean of the metal transit index (170), the coefficient in column (5) implies an increase of 0.005 ×170 = 0.85 percentage points, or about 52 percent of the total change.

Table 2: Metal transit trade and archaeological sites. Difference-in-differences (Stone Age vs Bronze Age)

	Any archaeological site (×100)					
	OLS		2SLS IV			
	(1)	(2)	(3)	(4)	(5)	(6)
	Panel A: Second stage Bahn (2000)					
Metal transit index (IHS) × Bronze Age	1.104	1.485	0.967	1.667		
	(0.252)	(0.263)	(0.419)	(0.715)		
	[0.262]	[0.251]	[0.466]	[0.745]		
Metal transit index × Bronze Age					0.005	0.012
					(0.002)	(0.005)
					[0.003]	[0.006]
	Panel B: Second stage Pleiades Gazetteer (Bagnall, 2022)					
Metal transit index (IHS) × Bronze Age	0.552	0.533	0.546	0.473		
	(0.101)	(0.110)	(0.124)	(0.226)		
	[0.114]	[0.119]	[0.129]	[0.219]		
Metal transit index × Bronze Age					0.003	0.003
					(0.001)	(0.002)
					[0.001]	[0.001]
	Panel C: First stage					
Natural metal transit index (IHS) × Bronze Age			0.490	0.308	86.183	43.091
			(0.031)	(0.030)	(13.032)	(8.274)
			[0.036]	[0.034]	[13.016]	[7.986]
Geography controls×Bronze Age	No	Yes	No	Yes	No	Yes
Grid cell fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Continent×Bronze Age	Yes	Yes	Yes	Yes	Yes	Yes
Observations	21,940	21,940	21,940	21,940	21,940	21,940
Mean dependent variable Panel A	4.913	4.913	4.913	4.913	4.913	4.913
Mean dependent variable Panel B	1.759	1.759	1.759	1.759	1.759	1.759
R-squared Panel A	0.615	0.618				
R-squared Panel B	0.823	0.825				
First-stage F-stat (5×5 grids)			251.000	102.289	43.734	27.122
First-stage F-stat (Conley 1000km)			184.831	82.079	43.400	28.807

Notes: This table reports difference-in-differences OLS and 2SLS IV estimates of equation (6). The unit of observation is a 1×1 grid-cell k in period p (Stone Age or Bronze Age). 'Any Site' equals 100 rather than 0 for a grid cell k in period p if there is at least one archaeological site dating from period p and located in grid cell k. 'Metal transit index' is defined by equation (2), while 'Natural metal transit index' is defined by equation (3). Geography controls include proximity to mines, proximity to croplands, sea dummy, river dummy, mountains dummy, and centrality in the transport network. Panel A uses archaeological data from Bahn (2000), while Panel B uses data from Bagnall (2022). Panel C reports first-stage estimates (equation (7)). Standard errors clustered at the level of 5×5 degree grid cells are reported in parentheses. Standard errors computed using the approach of Conley (1999) (cut-off 1000km) are reported in square brackets.

Panel (a)



Panel (b)

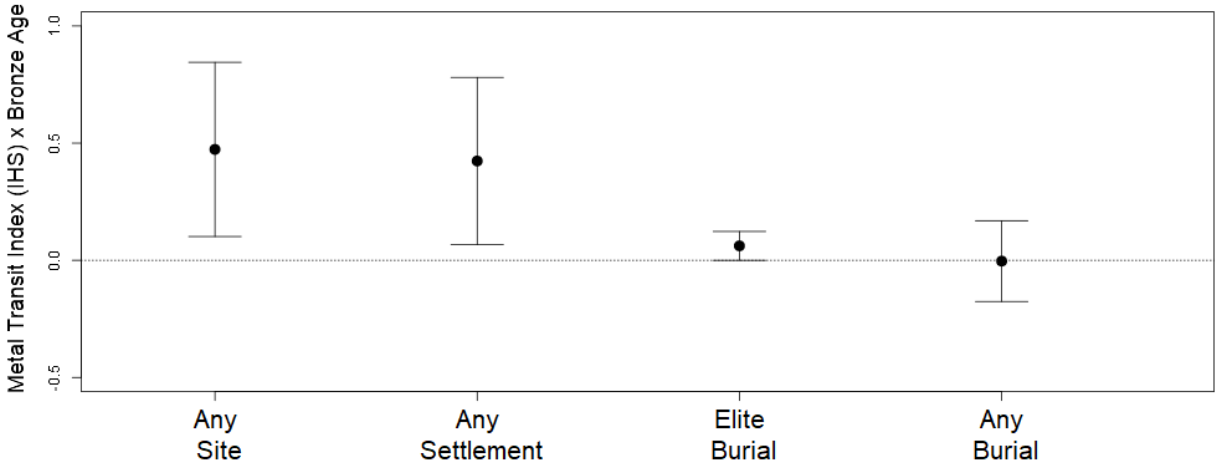


Figure 2: Estimated impact of the metal transit index across different types of archaeological finds

Figure depicts 2SLS IV point estimates and 90% confidence intervals of β in equation (6). Standard errors are clustered at the level of 5×5 degree grid cells. Dependent variable equals 100 if an archaeological site is present in a given 1×1 grid cell k dating from period p and 0 otherwise. Panel (a) uses data on archaeological sites from Bahn (2000) and panel (b) uses data from Bagnall (2022).

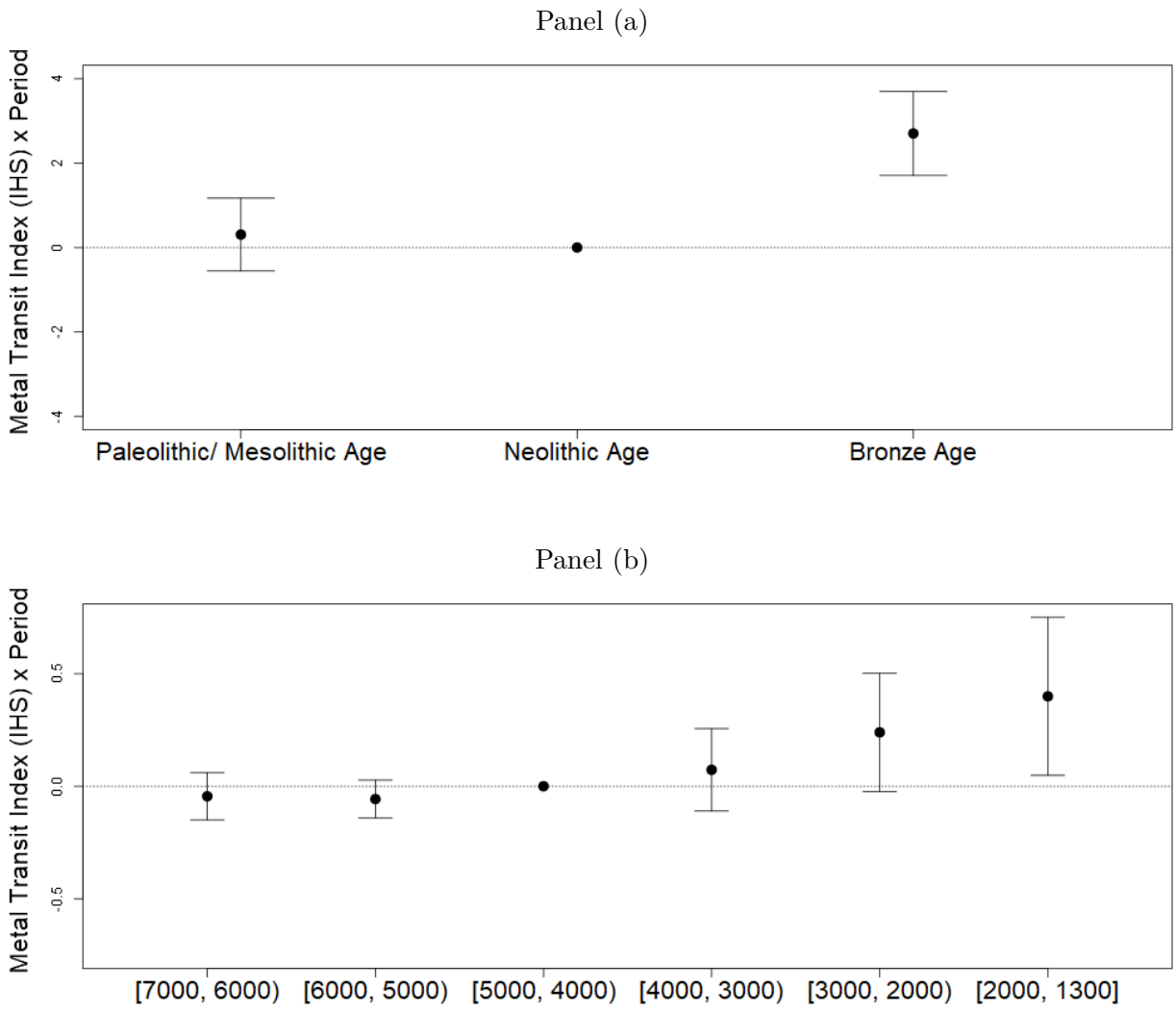


Figure 3: Estimated impact of the metal transit index on the spatial location of archaeological sites over time
Figure depicts 2SLS IV point estimates and 90% confidence intervals of ψ_p in equation (8). Standard errors are clustered at the level of 5×5 degree grid cells. Dependent variable equals 100 if an archaeological site is present in a given 1×1 grid cell k dating from period p and 0 otherwise. Panel (a) uses data on archaeological sites from Bahn (2000); the reference period is the Neolithic Age. Panel (b) uses data from Bagnall (2022); the reference period is the time span $[5000, 4000)$ BC. See Section 2 for details on definition and construction of variables.

We conducted a series of robustness checks in the same spirit as those outlined in the previous subsection. Full results are presented and discussed in Online Appendices D.2 and D.4. Specifically, we assess the robustness of our findings to: (1) addressing the potential endogeneity in the relative transportation costs across different modes of transportation estimated for the Bronze Age; (2) employing alternative standard error specifications; (3) redefining the unit of observation using 0.5×0.5 degree grid cells; and (4) controlling for non-metal transit trade in the regressions.

2. Warfare in archaeological data

In Online Appendix D.5, we explore the relationship between metal transit trade and warfare. We systematically searched the online collections of six major museums for weapons dating to the pre-Bronze Age and Bronze Age. Following Kristiansen and Horn (2018), we treat the presence of weapons as a proxy for organized military activity. Based on this dataset, we re-estimate regression equation (6) using 2SLS, with an indicator for the presence of weapons as the dependent variable. The estimates show a significant shift in this archaeological proxy for warfare toward regions situated along metal transit corridors following the advent of metallurgy. This result is consistent with a substantial archaeological and historical literature which emphasizes that one of the key drivers of conflict and expansionist campaigns in the Bronze Age was the desire to control strategic nodes within regional and interregional trade systems.[23] For example, in a regional analysis of weapon distribution in Central Europe, Kristiansen and Horn (2018) conclude: *"warriors played an essential role because they had many functions during the Bronze Age. The most important was to control trade and its valuables – from metal to amber and salt. Every community needed it."* They further note that: *"organized trade made warriors indispensable, and warfare could take on huge proportions when it came to the control of trade routes, or rather their most important hubs/bottlenecks, such as Bernstorf".*

4.3 China

Bronze metallurgy in China originated around 2000–1900 BC and likely developed independently within China, without outside influence. The archaeological site of Erlitou is the largest among the sites dating from these years; it is located at the center of an intricate network of rivers, connecting metal-rich mountains to the fertile Chinese lowlands. Recent archaeological literature argues that Erlitou took advantage of this strategic position to direct the metal trade in the region, eventually becoming the capital of a trade civilization (and probably the first state in Chinese history). We provide a detailed review of this literature in Online Appendix E.4. In this subsection, we show that the link between trade and the Urban Revolution in China extends beyond the Erlitou civilization.

One major advantage of focusing on China is the availability of a comprehensive catalog of Chinese archaeological sites. For each site, we record the presence of six key archaeological markers of the Urban Revolution: 1) large settlements, 2) monumental buildings, 3) ruling class exempt from manual tasks, 4) a standardized system of measures and recordings used in exchange and production, 5) writing, 6) highly developed art forms. We then estimate the difference-in-differences model described in equation (6), assuming the Bronze Age in China begins around 2070 BC—traditionally associated with the Xia Dynasty—and ends in 771 BC with the Western Zhou Dynasty.

The main results are presented in Table 3. We use two outcome measures to capture urbanism

[23] A classic example is the Trojan War, which several scholars have linked to Troy's control of the Dardanelles—the most accessible crossing point between Europe and Asia (Rose, 1998; Cline, 2013, p. 47).

and statehood. The first is a binary indicator capturing whether a site exhibits at least one of the six markers we measure in the data. The second is the first principal component derived from these six indicators.

Table 3: Metal transit trade and archaeological sites in China. Difference-in-differences (Stone Age vs Bronze Age)

	OLS		2SLS IV			
	Any Criteria (×100)	Principal Component (SD)	Any Criteria (×100)	Principal Component (SD)	Any Criteria (×100)	Principal Component (SD)
	(1)	(2)	(3)	(4)	(5)	(6)
	Panel A: Second stage					
Metal transit index (IHS) × Bronze Age	4.684	0.140	7.080	0.199		
	(0.690)	(0.025)	(0.869)	(0.031)		
	[0.374]	[0.020]	[0.495]	[0.027]		
Metal transit index × Bronze Age					0.031	0.001
					(0.011)	(0.000)
					[0.007]	[0.000]
	Panel B: First stage					
Natural metal transit index (IHS) × Bronze Age			1.039	1.039	241.192	241.192
			(0.096)	(0.096)	(94.570)	(94.570)
			[0.090]	[0.090]	[79.239]	[79.239]
Geography controls×Bronze Age	No	No	No	No	No	No
Grid cell fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Continent×Bronze Age	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,138	2,138	2,138	2,138	2,138	2,138
Mean dependent variable	17.914	0	17.914	0	17.914	0
R-squared	0.761	0.725				
First-stage F-stat (5×5 grids)			116.036	116.036	6.504	6.504
First-stage F-stat (Conley 1000km)			132.038	132.038	9.164	9.164

Notes: Panel A reports difference-in-difference OLS (columns (1)–(2)) and 2SLS (columns (3)–(6)) estimates of equation (6), while panel B reports the first-stage estimates (equation (7)). The unit of observation is a 1×1 grid-cell k in period p (Stone Age or Bronze Age). 'Any Criteria' equals 100 (rather than 0) if there is any archaeological site in grid cell k dating from period p that displays some evidence of at least one of the following features: large settlement, monumental building, a ruling class exempt from manual tasks, a system of recording, writing, art. 'Principal Component (SD)' is the standardized principal component. 'Metal transit index' is defined by equation (2), while 'Natural metal transit index' is defined by equation (3). Geography controls include proximity to mines, proximity to croplands, sea dummy, river dummy, mountains dummy, and centrality in the transport network. Standard errors clustered at the level of 5×5 degree grid cells are reported in parentheses. Standard errors computed using the approach of Conley (1999) (cut-off 1000km) are reported in square brackets.

Columns (1) through (4) present both OLS and 2SLS estimates. Across all specifications, we find a strong and statistically significant association between the metal transit index and the increase in Urban Revolution markers during the Bronze Age, relative to earlier periods. Notably, the 2SLS coefficients are roughly twice as large as their OLS counterparts. We conjecture that this gap may reflect particularly severe measurement error in the construction of the metal transit index for China.[24] In columns (5) and (6), we replace the IHS-transformed regressor with the untransformed metal transit index. These estimates should be interpreted with caution, as the first-stage F-statistics fall below the conventional threshold of 10, indicating potential weak instrument concerns. With this caveat in mind, the estimate in column (5) implies that increasing the metal transit index from zero to its mean value (170) raises the probability of observing at least one Urban Revolution marker during the Bronze Age by approximately 5 percentage points-relative to a baseline mean of 18 percent.

Figure 4 presents two additional exercises. Panel (a) reports 2SLS difference-in-differences estimates for each of the six Urban Revolution markers separately, including the full set of controls.

[24]Bronze Age mining data in China are likely to be especially noisy and incomplete relative to Europe and the Middle East. For example, Jie et al. (2024) observe: "Chinese scholars have hitherto focused mainly on the construction of regional cultural types and the chronological framework [...] In terms of the research on metallurgical technology, excavated metal artifacts have been the primary focus, with little attention paid to mining and processing sites."

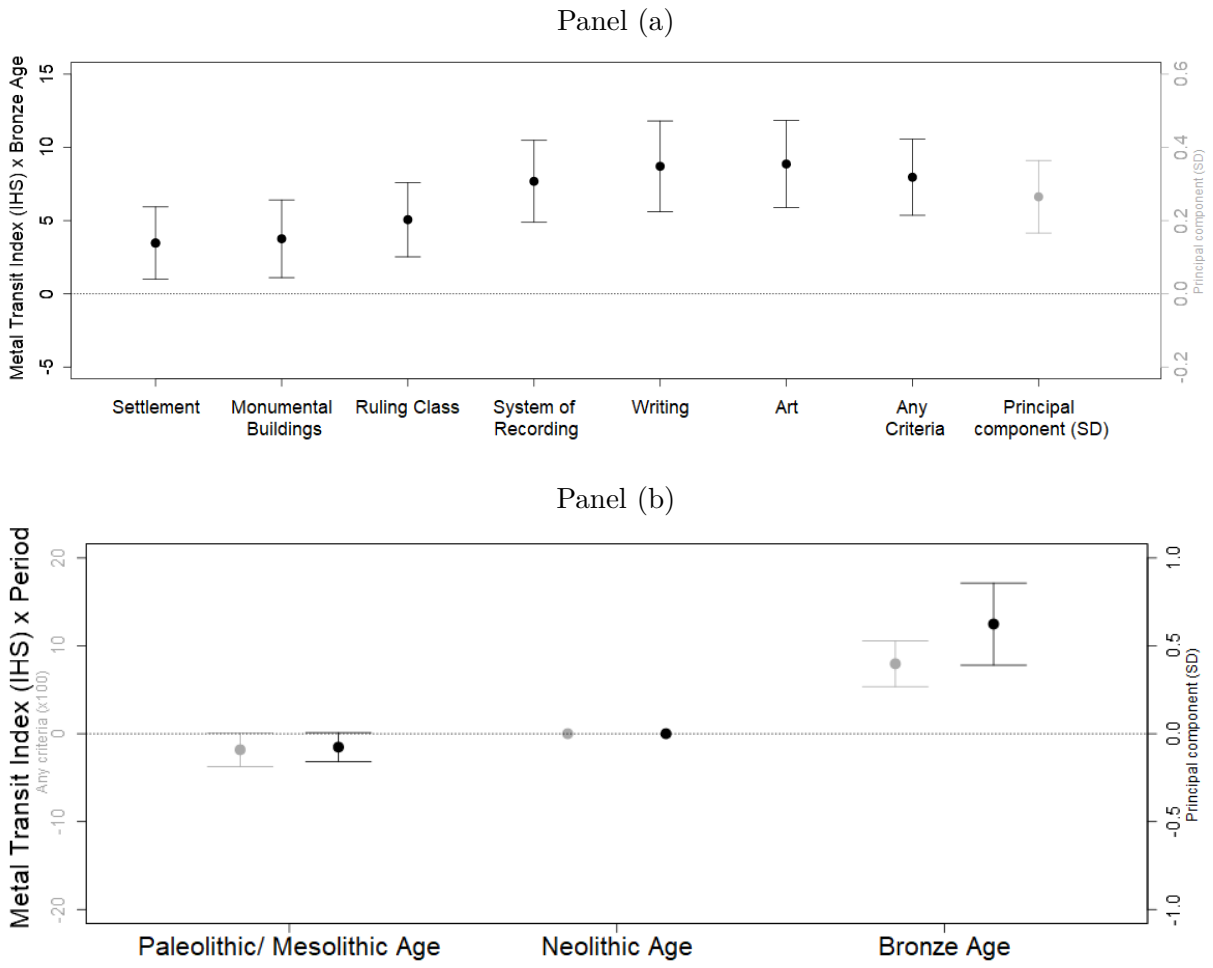


Figure 4: Estimated impact of the metal transit index on the rise of the Urban Revolution in China

Panel (a) depicts 2SLS IV point estimates of β in equation (6) and 90% confidence intervals. Each data point corresponds to a regression with a different dependent variable. The dependent variable equals 100 (rather than 0) if there is any archaeological site in 1×1 degree grid cell k dating from period p (Stone Age or Bronze Age) that displays some evidence of: large settlements, monumental buildings, a ruling class exempt from manual tasks, a system of recording used in the production process, writing, figurative art. 'Any criteria' equals 100 (rather than 0) if there is any of the six features present in grid cell k dating from period p. Grey: Dependent variable is the (standardized) first principal component obtained from pooling these six features. Standard errors are clustered at the level of 5×5 degree grid cells. Panel (b) depicts 2SLS IV point estimates of ψ_p in equation (8) and 90% confidence intervals. Standard errors are clustered at the level of 5×5 degree grid cells. Grey: Dependent variable equals 100 (rather than 0) if at least one of the following features is present in sites in 1×1 degree grid cell k dating from period p: large settlements, monumental buildings, evidence of a ruling class exempt from manual tasks, a system of recording used in the production process, writing, figurative art. Black: Dependent variable is the (standardized) first principal component obtained from pooling these six features. The reference period is the Neolithic.

In all cases, the estimated coefficients are positive and statistically significant. Panel (b) shows the dynamic version of the model described in equation (8), using the Neolithic as the reference category. We find no evidence of pre-trends. The association between the metal transit index and the presence of Urban Revolution markers emerges only during the Bronze Age.

4.4 Europe

In Online Appendix D.6, we investigate the relationship between trade routes and the rise of social hierarchies during the Early-Middle Bronze Age (3000-1300 BC) and the Late Bronze Age (1300-750 BC) in Western and Central Europe. While this region did not witness the emergence of full-fledged political states during the Bronze Age, it did experience a transition from classless bands and tribes to complex chiefdoms. A key advantage of focusing on this region is that the European Bronze Age is one of the most extensively studied periods in world archaeology. Its rich archaeological record allows us to trace both changes in metal transit routes—driven by the discovery of new mines and the depletion of old ones—and shifts in the location of the more complex hierarchies, as captured by elite burials.

Using a two-period panel dataset (covering the Early-Middle and Late Bronze Age) across 26 European countries, we estimate how changes in the metal transit index between these two periods relate to changes in the distribution of elite burials. The results indicate that increased metal transit activity is significantly associated with the emergence of elite burials, suggesting a robust link between metal transit trade and early forms of centralized leadership even within the European context.

5 Appropriability mechanism

The results so far suggest that trade in bronze inputs was likely an important driver of the Urban Revolution. Cities and states were more likely to emerge in regions situated along the metal trade network. In this section, we examine one potential mechanism underlying this relationship.

In Appendix D.8.1, we present a simple model inspired by Mayshar, Moav and Pascali (2022) extended to incorporate a spatial dimension. As in our empirical setup, space is structured into grid cells, with transport costs between adjacent cells serving as key exogenous parameters. Traders move metal from mines to farming regions, choosing routes that minimize total trade costs—defined as the sum of transport costs and expropriation rates. The level of expropriation in each cell depends on its political organization: each cell can be governed either as a hierarchy, in which a king establishes a monopoly on violence by paying a fixed cost and employs tax collectors, or as anarchy, where unorganized roving bandits operate. The model delivers three key insights. First, expropriation (or tax) rates are lower under hierarchy than under anarchy. This is because in anarchy, individual bandits do not internalize the negative externalities their theft imposes on others. Each maximizes their own gain without regard for the fact that excessive expropriation rates may cause traders to divert to other cells. In contrast, a centralized authority sets taxes to maximize overall revenue, which leads to more moderate expropriation rates. Second, total expropriation in a cell cannot exceed its "blockage cost"—the increase in transportation costs that traders would face if that cell were removed from the transportation network. Third, hierarchy Pareto-dominates anarchy: tax collectors are better off than bandits, and traders face lower expropriation.[25]

[25]This result follows a logic akin to the classic hold-up problem in the industrial organization literature, where the inability to

Taken together, these insights suggest that hierarchy is likely to emerge in cells with sufficiently high blockage cost—high enough to ensure that potential tax revenue exceeds the fixed cost of establishing a centralized rule. This prediction mirrors the conjecture that an extensive anthropological and archaeological literature has advanced to explain the rise of complex hierarchies during the Bronze Age. In the words of Earle et al. (2015): "In simple terms, the Bronze Age witnessed an emergence of social stratification based on control over commodity flow. [...] bottlenecks emerged in metal flows to offer emerging elites opportunities to extract surpluses."

In order to translate the insights of the model to the data, we compute, for each grid cell in the Old World, a modified efficiency centrality measure (Wang, Wang and Deng, 2019) in the metal trade network (see Appendix D.8 for details). This centrality measure captures the increase in global transport costs that would result if the cell were blocked to transit—that is, if no path connecting cropland with mines were allowed to intersect it. We refer to this centrality measure as the cell's blockage cost. The geographic distribution of these blockage cost, visualized in Figure 5, identifies the bottlenecks in the metal trade transportation network and, according to our theory, the locations with the greatest potential for the emergence of a local tax-extracting elite organized as a centralized state. Online Appendix D.8.3 provides further details on the correlation between the metal transit index and the blockage cost measure, including specific illustrative examples.

To assess the extent to which the rise of a tax-extracting elite explains our empirical findings on the relationship between the metal transit index and archaeological markers of the Urban Revolution, we re-estimate all benchmark OLS specifications from the previous section, now including both the metal transit index and the blockage cost as explanatory variables. If taxation plays a role, we expect the coefficient on the blockage cost to be positive, statistically significant, and sizable. Table 4 presents the results of these horse race regressions. Panel A reports estimates using the IHS-transformed metal transit index and blockage cost, while Panel B reports results using standardized versions of these variables. Standardization facilitates direct comparison of coefficient magnitudes. We focus the discussion on Panel B, although the findings in Panel A are qualitatively similar.

Column (1) reproduces the baseline OLS estimate from equation (4) with full controls; the estimated coefficient on the metal transit index is 0.732. In column (2), when both the metal transit index and blockage cost are included, the estimated coefficient on the metal transit index falls to zero, while the estimated coefficient on the blockage cost is positive, large, and statistically significant. This pattern is consistent with the hypothesis that the emergence of a trade-taxing elite helps explain the observed relationship between metal transit trade and the spatial distribution of Bronze Age cities. The results are robust to using probit estimation, as shown in columns (3) and (4). In columns (5) and (6), we estimate the OLS difference-in-differences regression using the Bahn (2000) panel; columns (7) and (8) replicate the analysis using the Atlas of Chinese Cultural Relics. In both datasets, the estimated coefficient on the interaction between blockage cost and the Bronze Age dummy is positive, large, and statistically significant in most specifications. Meanwhile, the

commit to future terms of exchange leads to inefficiencies. Our model is related to insights of Shleifer and Vishny (1993), who apply this logic to the context of corruption. Specifically, their article analyzes corruption under three institutional configurations: (1) a centralized bribe-collecting monopolist, (2) multiple uncoordinated monopolists each setting their bribe on complementary public goods, and (3) a competitive environment with no monopolists. In our framework, expropriation rates of transit trade function analogously to the bribes examined in their model, and the first two configurations map closely to what we describe in our model as the "state" and "anarchy" scenarios, respectively. Our model adds three additional elements: (i) the possibility of bypassing a given location (grid cell) instead of paying the tax or bribe; (ii) a fixed cost of organizing a monopoly over taxation, which endogenizes the transition from anarchy to a state; and (iii) an explicitly spatial dimension that reflects trade routes and geography.

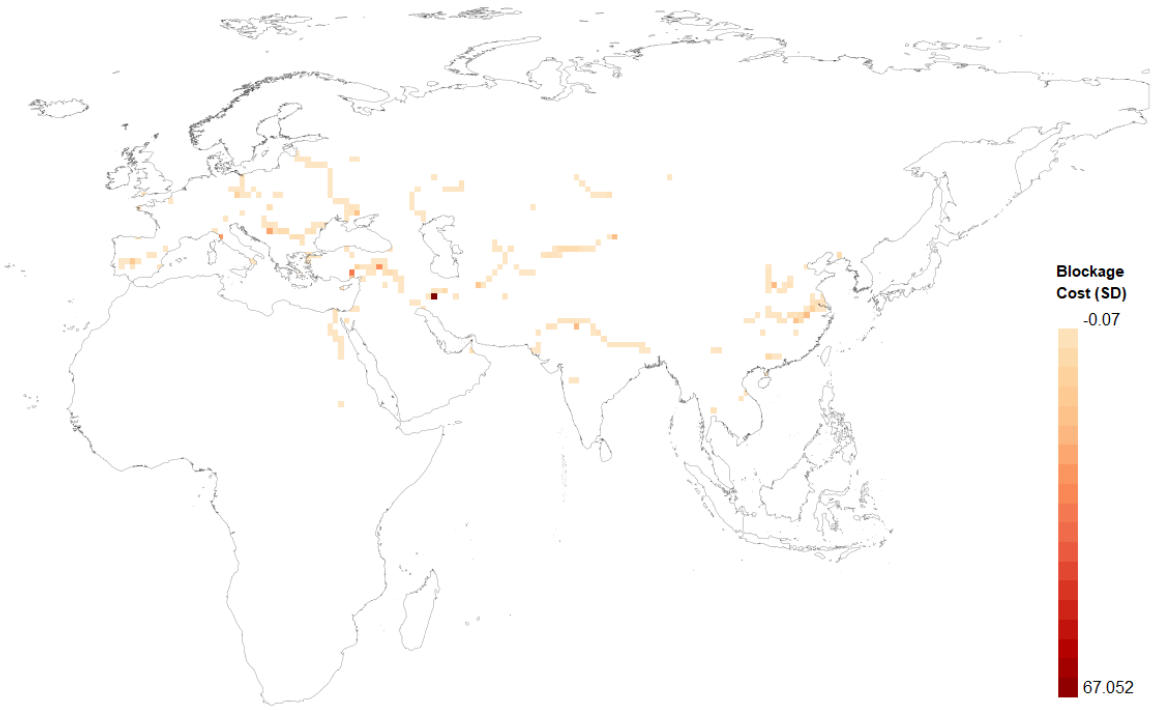


Figure 5: Blockage cost
Figure depicts the global cost of blocking a cell for transit trade in 3000 BC (see Appendix B for more details). Darker shadings imply greater costs.

corresponding coefficient on the interaction between the metal transit index and the Bronze Age dummy declines by 35 percent and 79 percent, respectively, when blockage cost is included as a regressor.

Table 4: Horse race: Metal transit index vs blockage cost

	Any city or citadel (×100) Green et al. (2024)				Any Site (×100) Bahn (2000)		Any Criteria (×100) Atlas of Chinese Relics	
	OLS		Probit		OLS		OLS	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: IHS-transformed explanatory variables								
Metal transit index (IHS)	0.353	0.073	0.086	0.048				
	(0.076)	(0.065)	(0.028)	(0.034)				
	[0.097]	[0.079]	[0.012]	[0.015]				
Blockage cost (IHS)		0.931		0.095				
		(0.264)		(0.039)				
		[0.306]		[0.019]				
Metal transit index (IHS) × Bronze Age					1.485	0.963	3.669	3.040
					(0.263)	(0.307)	(0.772)	(0.710)
					[0.251]	[0.267]	[0.412]	[0.449]
Blockage cost × Bronze Age (IHS)						1.736		2.033
						(0.649)		(1.483)
						[0.427]		[0.955]
Panel B: Standardized explanatory variables								
Metal transit index	0.731	-0.095	0.028	-0.051				
	(0.188)	(0.261)	(0.014)	(0.028)				
	[0.224]	[0.279]	[0.006]	[0.012]				
Blockage cost		1.183		0.139				
		(0.406)		(0.035)				
		[0.427]		[0.015]				
Metal transit index × Bronze Age					2.188	1.431	3.434	0.736
					(0.394)	(0.468)	(2.312)	(2.887)
					[0.327]	[0.344]	[1.269]	[1.486]
Blockage cost × Bronze Age						1.087		3.436
						(0.506)		(1.825)
						[0.364]		[1.337]
Geography controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Grid cell fixed effects	No	No	No	No	Yes	Yes	Yes	Yes
Continent×period fixed effects	No	No	No	No	Yes	Yes	Yes	Yes
Observations	10,970	10,970	10,970	10,970	21,940	21,940	2,138	2,138
R-squared Panel A	0.018	0.026	0.259	0.266	0.618	0.619	0.766	0.766
R-squared Panel B	0.017	0.029	0.239	0.265	0.617	0.618	0.756	0.757
Mean dependent variable	0.483	0.483	0.483	0.483	4.913	4.913	17.914	17.914

Notes: This table reports probit and OLS estimates. 'Metal transit index' is defined by equation (2); Appendix B describes the construction of the variable 'Blockage cost'. The dependent variables and the 'Geography controls' are the same as in Tables 1 (for columns 1–4), Table 2 (for columns 5–8). Standard errors clustered at the level of 5×5 degree grid cells are reported in parentheses. Standard errors computed using the approach of Conley (1999) (cut-off 1000km) are reported in square brackets. In Panel A, the metal transit index and blockage cost are transformed using the hyperbolic sine transformation. In Panel B, the metal transit index and blockage cost are standardized to a mean of zero and a standard deviation of one. Blockage cost is winsorized at the 1% level.

Caveats

It is important to note that these horse race regressions can only provide suggestive evidence for the relevance of an appropriability mechanism linking metal transit trade to the emergence of hierarchy. Three important caveats should be considered. First, since we cannot identify exogenous variations in the blockage cost variable, we acknowledge that these OLS estimates fall short of

establishing a causal relationship. Second, as is typical in OLS horse race regressions, discrepancies in measurement errors between the transit index and blockage cost variables may account for significant variations in the magnitude and statistical significance of the estimated coefficients on these two variables. Third, even if we were able to accurately identify the causal impact of both the transit index and blockage cost, it is plausible to conceive of models wherein higher blockage costs are associated with increased transit trade for the same level of our transit index.[26]

Appropriability and other mechanisms

Taken together, the results in Table 4 provide suggestive evidence that the emergence of a trade-taxing elite may have played a significant role in triggering the onset of the Urban Revolution. This finding complements other empirical studies in comparative development studying the link between geography and early urban and state development. In Online Appendix D.7, we conduct a Shapley decomposition to assess the relative explanatory power of our proposed mechanism compared to four complementary explanations identified in prior work: the emergence of farming surplus, the potential for irrigation, the rise of conflict in circumscribed environments, and the appropriability of cereal grains.

We estimate a cross-sectional probit model in which the presence of a city within a grid cell is regressed on a set of continent dummies, our measure of blockage cost, and proxies—mostly drawn from prior studies—to capture these four complementary explanations: cropland area in 3000 BC, increases in land productivity generated by irrigation (Bentzen, Kaarsen and Wingender, 2017), distance to cereal domestication regions (Mayshar, Moav and Pascali, 2022), and a geographic circumscription index (Schönholzer and François, 2023). Each of these four proxies has a relatively high Shapley value, ranging from 0.03 to 0.06, while the continent dummies have a value of 0.03. Notably, the Shapley value for the blockage cost measure is 0.08—the highest among all variables considered. While the caveat remains that we cannot rely on an exogenous instrument for the blockage cost, it is nevertheless reassuring that including the blockage cost in a regression of Bronze Age city location on commonly studied geographic determinants leads to a substantial increase in the model's R^2.

6 Case studies

This section presents a series of case studies tracing the relationship between the expansion of long-distance trade and the emergence of city-states in Mesopotamia during the Early and Middle Bronze Age. Online Appendix E provides complementary case studies focusing on analogous processes in the Indus Valley, the city of Mari, the Aegean, and China. We draw on a broad literature in archaeology, history, and anthropology that has explored how the growth of interregional trade contributed to the rise of the earliest urban centers and state structures. Four broad empirical regularities appear to be common across these early civilizations. First, Bronze Age urban centers are consistently located at strategic nodes within regional and interregional trade networks, particularly at logistical chokepoints, overland corridors, and navigable river junctions. Second, a substantial body of archaeological research has linked the initial urbanization of these societies to

[26] For instance, suppose we extend our theoretical framework to include congestion costs. In such a setting, bottlenecks might attract disproportionate volumes of trade despite experiencing higher congestion—simply because of the lack of alternatives. In this case, our blockage cost measure could partly reflect trade intensity rather than fiscal appropriability. As a result, in the horse race regressions, the blockage cost coefficient would capture both the scope for taxation and the underlying scale of trade, even after conditioning on the metal transit index.

their integration into long-distance transit trade, especially in metals, and documents that early urban elites were directly involved in the control and exchange of metal commodities. Third, recent work in archaeology and historical analysis emphasizes that the expansion of urban cores and the formation of early imperial polities were frequently motivated by the desire to secure trade corridors and gain direct access to scarce resources, especially metal ores and mining regions. Fourth, the revenues derived from taxing long-distance transit trade—particularly in high-value commodities like metals and aromatics—were likely substantial and may have funded a significant share of the emerging state apparatus, including specialized bureaucracies, monumental architecture, and other forms of public infrastructure. To visually link the empirical strategy presented in the preceding sections with the case study evidence, Figure 6 displays a composite map. It overlays the computed least-cost paths connecting cropland to mines, the estimated blockage cost of each grid cell, the spatial distribution of Bronze Age cities from our dataset, and key archaeological sites and regions discussed in the case studies.

6.1 The Uruk Period (4000-3100 BC)

Uruk has long been considered the first true city in human history, not only for its large population but also for its early development of complex administrative systems, protowriting, and the presence of separate residential, industrial, and administrative sectors. Situated along a now-dried branch of the Euphrates River, it was deeply integrated into southern Mesopotamia's riverine transport networks, which facilitated both regional integration and long-distance exchange. The precise origins of Uruk remain a subject of debate, but a long-standing tradition in archaeology interprets the so-called Uruk expansion—the diffusion of southern Mesopotamian institutions and urban settlements into northern Mesopotamia, Syria, western Iran, and parts of Anatolia during the late 4th millennium BC—as a process that *"was at least partially motivated by the desire to channel raw materials, especially metals, stones, and probably also timber from the surrounding highlands"* (Potts, 1993). The alluvium itself lacked mineral resources but was strategically situated between resource-rich highland zones and fertile agricultural lowlands. Uruk's expansion—an outward spread of southern Mesopotamian settlements and institutions into neighboring regions in the late 4th millennium BC—suggests a calculated effort to structure and maintain transit routes through which critical resources, especially copper, could be directed toward population centers. Archaeological evidence attests to this expansion in several interlinked forms:

(1) The establishment of large, fortified enclaves at key logistical junctures—sites like Carchemish, Habuba Kabira, Tell Brak, Samsat, and Nineveh—which sat at intersections of major east–west overland routes and north–south watercourses. In the words of Algaze (1989): *"Although the riverine locations of the enclaves would seem to suggest that a crucial factor determining their placement was control of the north-south waterways, a detailed analysis reveals that they are also oriented along east-west overland routes of communication across Syro-Mesopotamia known from classical times. Each, it seems, commands a historical juncture where the principal overland routes intersect the rivers."*

(2) Smaller intermediary stations, likely functioning as logistical relays or depots between the southern core and peripheral enclaves, further evidence the infrastructural nature of the expansion.

(3) Remote outposts in intermontane valleys, such as Godin Tepe and Tepe Sialk, which appear to have been strategically located to provide access to highland metal resources, particularly copper.

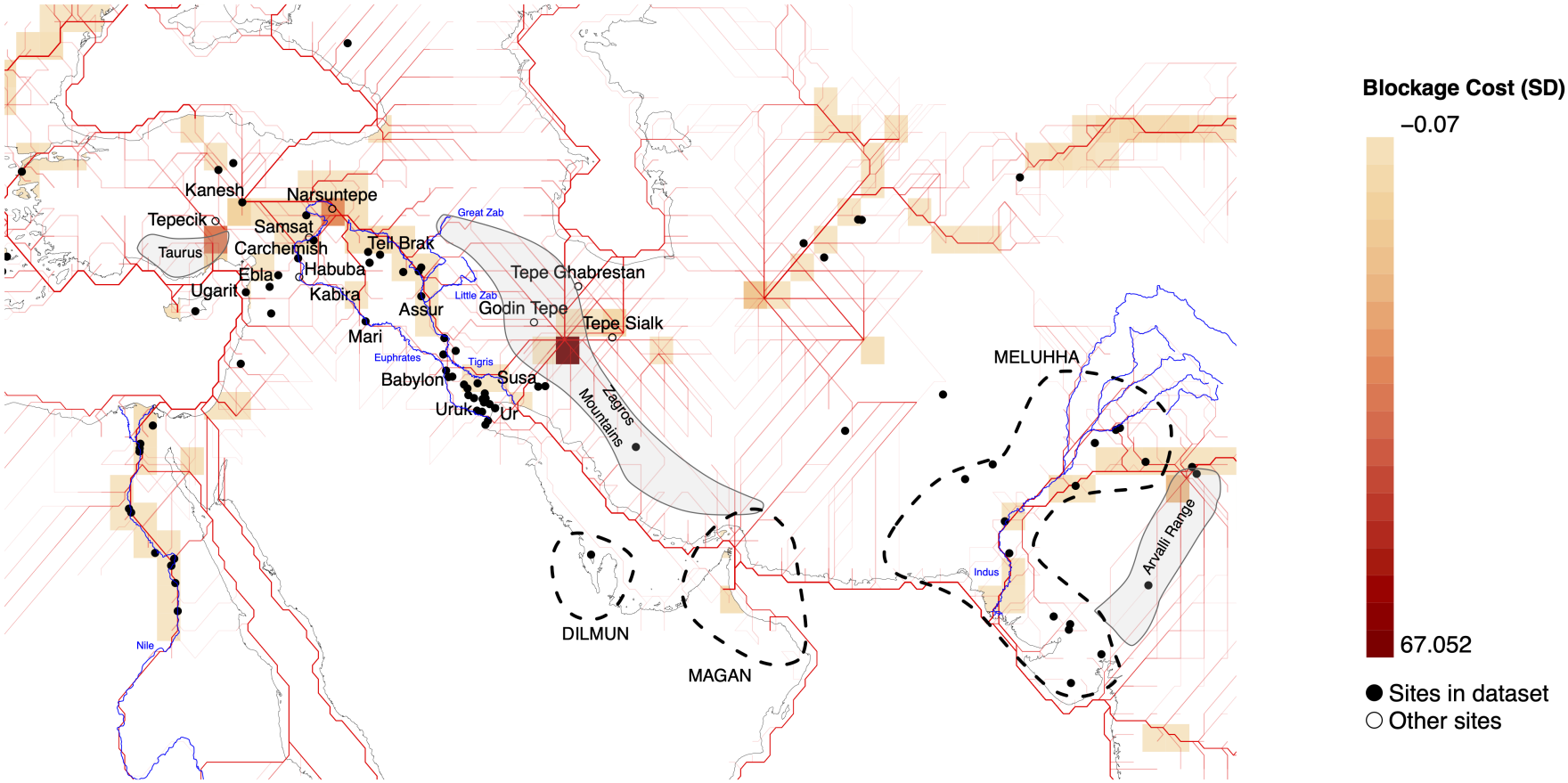


Figure 6: Transit paths, blockage cost, and cities in Mesopotamia

The figure shows the locations of Bronze Age cities (circles), least-cost paths connecting mines to croplands (red lines), and blockage costs (red shading). Also shown are the primary rivers (blue lines) and mountain ranges that were home to important metal deposits (gray shaded areas). Solid circles represent cities included in our dataset (Green et al., 2024; Reba, Reitsma and Seto, 2016), while empty circles represent additional sites mentioned in the case study.

These patterns are reinforced by the distribution of Uruk ceramics and administrative tools at indigenous metallurgical centers in copper-rich areas. In the Taurus mountains (Turkey), for example, Uruk pottery has been found at Tepecik, a site adjacent to the Ergani copper district, and at Norsuntepe, both key nodes of prehistoric copper production. On the Iranian plateau, Uruk materials appear at centers like Tepe Ghabrestan, close to the rich ore fields of Anarak and Veshnoveh. These sites had been active in copper metallurgy since the 5th millennium BC and were connected to Mesopotamia via established overland corridors, including the Zagros passes, the Solduz Valley, and the Lesser Zab. The distribution of Uruk ceramics in such contexts—sometimes minimal but archaeologically diagnostic—suggests not colonization but targeted commercial or logistical presence, likely oriented toward facilitating the extraction or exchange of metals.

Taken together, these patterns point to a coherent strategy of resource-oriented expansion, in which an important factor driving Uruk urbanism was the need to intercept and redirect the movement of highland commodities, especially copper, toward fertile agricultural land with dense populations.

6.2 The Early Dynastic Period (2900–2334 BC)

The Early Dynastic period marks a pivotal phase in Mesopotamian history, characterized by the emergence of independent city-states, each governed by its own ruler. This era provides the first clear evidence of dynastic kingship, inter-city warfare, and diplomatic interaction, alongside the appearance of legal, historical, and administrative texts. By this point, all of the core elements of the Urban Revolution, as outlined by Gordon Childe—including social stratification, writing, urban planning, surplus mobilization, and state institutions—are fully present. It is also the first period in which written records and archaeological data converge, offering a detailed and coherent picture of long-distance trade networks and early state formation.

Technologically, this period witnessed a shift from pure copper to arsenical bronze, which began to appear in tools, weapons, and ornaments from ca. 2900 BC. However, it is not until ca. 2600 BC that we observe clear evidence for the mass production of arsenical bronze in specialized urban workshops. Around the same time, the earliest confirmed examples of tin bronze appear, although initially limited to elite contexts such as funerary assemblages and prestige goods. The broader diffusion of tin bronze was gradual: it became standardized in administrative and military contexts only from ca. 2300 BC, as indicated by inventories and weapon caches.[27]

By this stage, Mesopotamian city-states had become central nodes in an expansive transregional trade network, primarily oriented around the exchange of metals and textiles. This network connected the eastern Mediterranean and the Nile Valley with the Persian Gulf, the Arabian Peninsula, the Iranian plateau, and the Indus Valley. As Foster (2015) observes: *"The principal commodities referred to in pre-Akkadian commercial documents were copper, tin, and tin-bronze, as well as conifer woods and the aromatics made from them, which were purchased with wool, textiles, and silver."*

Copper entered Mesopotamia through two principal routes. The first was overland, linking the ore-rich regions of the Zagros (Iran) and Taurus mountains to the southern plains via the water systems of the Tigris and Euphrates (Muhly, 1973; Morr et al., 2013). The second was maritime, connecting the ports of Ur and other southern cities through the Persian Gulf to the copper-producing regions of

[27] Widespread use of tin bronze in common tools and utilitarian objects does not become archaeologically visible until the final century of the third millennium BC.

Magan (Oman), Dilmun (Bahrain), and Meluhha (likely the Indus Valley).[28] Lead isotope analyses further suggest that some copper imported through this network originated from the Aravalli Hills in southern Rajasthan (Begemann and Schmitt-Strecker, 2008).

In contrast, the sources of tin in Bronze Age Mesopotamia are less clearly defined. Cuneiform texts suggest that the principal sources were located to the east, with western Afghanistan and the Zagros mountains being the most probable candidates. Tin from Afghanistan likely reached Mesopotamia via maritime routes through Magan (Cleuziou, 1982), while a land route from the Iranian plateau appears to have developed somewhat later. There is also limited but notable evidence of Bronze Age tin mining in Anatolia (e.g., Göltepe), which has led to the hypothesis that some of the tin used in Mesopotamia may have also derived from Anatolian sources (Yener and Vandiver, 1993; Yener, 2021).

As illustrated in Figure 6, all the major routes connecting copper and tin deposits with Mesopotamian cropland are captured by our estimated least-cost paths. In particular, note the bottleneck in the Taurus mountains (indicated by the darkest shading), which served as a critical transit corridor linking the fertile agricultural lowlands of Mesopotamia to metal-rich regions in the Taurus. The map also highlights the maritime route linking the Indus Valley to Mesopotamia via the intermediary hubs of Magan and Dilmun.

The Royal Archives of Ebla (ca. 2350 BC) offer the largest surviving assemblage of economic records from this period and provide an unparalleled view of the scale and organization of trade. Six key observations emerge from this corpus:

(1) Foreign trade and interregional contacts are the most frequently discussed topics in these texts.

(2) The commercial network centered on Ebla was vast, encompassing partners as far as Egypt, northern Syria, western Anatolia, the Balikh and Khabur Valleys, the Zagros foothills, and the Upper Euphrates, and extending as far south as Kish in northern Babylonia.

(3) The Euphrates River served as a commercial artery akin to the Silk Road of later eras. Trade along the Euphrates was primarily intermediary in nature, with little direct exchange between producers and final consumers. Instead, states and cities formed an interconnected chain of intermediaries, each profiting from and depending on the flow of goods along this route.

(4) The principal goods transiting through Ebla included metals—notably tin, silver, and gold—alongside precious stones such as lapis lazuli, timber, and a wide array of textiles. In contrast, Ebla's exported products were relatively limited, consisting mainly of olive oil and locally produced textiles, suggesting that the city's economic strength lay not in production but in its role as a commercial intermediary.

(5) There is evidence of the existence of tolls or fees that the various members of the Euphratean network collected for the passage of commercial goods through their territories. These levies—referred to in the sources as a "Euphrates toll" or "tithe"—appear to have been a formal and institutionalized mechanism of taxation. Importantly in the context of this paper, it has been argued that Ebla accumulated staggering quantities of precious metals through the imposition of

[28] Magan was the primary supplier of copper ore. Recent estimates suggest that as much as half of the copper consumed in Mesopotamia originated there (Giardino, 2019). Dilmun is referenced in archaic texts from Uruk as early as the late fourth millennium BC. The importance of Dilmun is reflected in the references to "Dilmun boats" and the use of the "Dilmun weight standard" in commercial transactions. However, epigraphic and archaeological evidence suggest that Dilmun functioned as a trading post, with the ultimate source of copper being Meluhha (Begemann et al., 2010), usually believed to correspond to the Indus Valley Civilization (Muhly, 1973).

customs duties on long-distance trade transiting its domain (Benati and Bonechi (2020); Marchesi (2013)). As Steinkeller (2021) concludes: *"It may be reasonably argued that [Ebla's] wealth derived primarily from her involvement in intermediary trade. I submit that because of her favorable geographical situation and her astute politics vis-à-vis the other dominant powers of the region, Ebla had been able to become, however briefly, the center of an astonishingly large interregional trade-network. This network [...] because of its size, could even be described as global."*

(6) The Ebla archives also highlight the prominence of Mari as the dominant regional power along the middle Euphrates during this period. Based on his analysis of the textual evidence preserved in these archives, Steinkeller (2021) argues that Mari's expansion was primarily commercial in character rather than territorial: *"Although I have repeatedly used the word "control" to describe Mari's dominance over the middle course of the Euphrates, it would be incorrect to think that this "control" involved territorial possession and the direct exploitation of natural resources, whatever they may have been. As far as our data permit to ascertain, Mari's sole objective in that regard was to maximise her profits from the Euphratean commerce. This she accomplished either directly, by establishing trading outposts at key traffic junctions or even by outright control of such places (and here the case of Tuttul comes to mind); or she did it indirectly, by forcing the various intermediaries to share their profits from duties on traded goods."*

6.3 City-states in the Old Assyrian trade system (1950-1750 BC)

Assur stands out as a compelling example of a city-state that prioritized long-distance trade as a core political and economic objective, deriving much of its revenue from taxing transit merchants operating within a flourishing interregional exchange system. Our understanding of its institutions is based primarily on a unique archive of approximately 23,500 cuneiform tablets, excavated from the site of Kanesh in central Anatolia.

Assur was located on a sandstone cliff overlooking the Tigris River. Although surrounded by relatively unproductive farmland, Assur's position at the confluence of the Tigris and its two major tributaries—the Great Zab and the Little Zab—made it a strategic transit point linking Anatolia, Sumer, and the Iranian plateau.

One of Assur's earliest rulers, Ilushuma I, left behind an inscription that directly connects the city's rise to the trade in metals:

"Ilushuma, vice-regent of Ashur, built the temple for the goddess Ishtar, his mistress, for his life. A new wall [...] I constructed and subdivided for my city house-plots. The god Ashur opened for me two springs in Mount Ebih and I made bricks for the wall by the two springs. [...] The freedom of the Akkadians and their children I established. I purified their copper. I established their freedom from the border of the marshes and Ur and Nippur, Awal and Kismar, Der of the god lshtaran, as far as the City (i.e., Assur)."

The inscription opens with a reference to urban foundations, then shifts to the economic geography underpinning Assur's rise. It alludes to three principal trade routes that brought copper to the city from southern Mesopotamia.[29] The centrality of metal commerce to Assur's economy is echoed in a later inscription from Ilushuma's successor, Erishum I, which records tax exemptions on trade in copper, tin, and barley transiting through the city.

[29] One from Ur, the main copper entry point from the Gulf, via Nippur; one along the Tigris from Awal and Kismar in the Hamrin region; and one from Elam via Der, east of the Tigris, passing through Susa (Larsen, 1976).

Between 1950 and 1750 BC, Assur came to dominate the trade corridor linking upper Mesopotamia with Anatolia, as well as a broader commercial circuit encompassing much of central Anatolia. These routes likely formed part of a chain of interlocking trade networks, which may have ultimately connected the Chinese frontier with the Balkans (Barjamovic, 2008). Within this system, Assur and Kanesh functioned as the principal commercial hubs for Assyrian merchants.

Assur served as the endpoint of what contemporary texts refer to as the "caravan of the Lower Country"—a trade route that brought tin, likely mined in Central Asia, from the city of Susa. Assyrian merchants acquired this tin—along with textiles—at Assur, transported it to Kanesh (the official clearinghouse for imports entering Anatolia), and from there redistributed it to other Anatolian settlements. In parallel, Kanesh also served as a central node in the Anatolian copper trade, connecting the copper-producing regions of northern Anatolia with urban centers in the west.

As shown in Figure 6, our estimated least-cost paths and blockage costs align well with the historical record. Both Assur and Kanesh exhibit high blockage costs, and the course of the least-cost paths accurately reflects Assur's function as a key node connecting southern Mesopotamia with Anatolia, and Kanesh's central position within the internal Anatolian trade network.

The Assyrian texts attest to an extraordinarily large-scale and sustained volume of trade.[30] They also depict a flourishing market economy based on free enterprise and private initiative, with profit-oriented and risk-taking merchants supported by sophisticated financial contracts and a well-functioning judicial system.

Assur established a network of trade outposts—or karums—in Anatolia's city-states. These settlements were equipped with legal and financial institutions that mirrored those of the mother city. Trade was tightly regulated through a series of tolls imposed at different stages of the commercial journey. Goods were sealed and accompanied by an official document (awītum), carried by the caravan leader and presented to inspectors en route. Toll taxes along the route from Assur to Kanesh were substantial and likely the main source of revenue for the Assyrian state. For instance, the value of tin would normally double between Assur and Kanesh, and Dercksen (2014) estimates that roughly one-fifth of this price increase was attributable to tolls and duties collected en route.[31]

After leaving Kanesh, Assyrian caravans encountered further toll taxes levied by local rulers of the Anatolian city-states. These fees represented a substantial source of revenue for these polities (Barjamovic, 2020).[32] Trade paths were often negotiated between the City Assembly (ālum) of Assur and these Anatolian authorities.[33] In exchange for these revenues, local rulers were expected to provide secure passage for merchants and bear liability for theft or injury. This arrangement presupposed clear definitions of political boundaries along trade roads and entailed state investment in infrastructure such as guard posts, inns, bridges, and roads. Standing contingents monitored rural

[30] Recent estimates suggest that approximately 1,500 donkeys, carrying around 15 tons of tin and 32,000 textiles annually, traveled from Assur to Kanesh (Stratford, 2010; Barjamovic, 2018). To put this into perspective: 15 tons of tin could be exchanged in Kanesh for an average of 2 tons of silver (prices fluctuated), a sum sufficient to cover the annual living costs of 2,000 to 4,000 people in Kanesh (Dercksen, 2014). The Assyrian texts also point toward large volumes of trade in textiles and copper. In one recorded instance, a single shipment included 23 tons of copper and 15 tons of wool. More broadly, references to the transport of copper by the ton are common in the archive (Barjamovic, 2011), highlighting the industrial scale of the Assyrian trade operation across the Anatolian landscape.

[31] Upon arrival in Assur, tin bound for Anatolia was wrapped in textiles and sealed—hence the term "sealed tin"—while "hand-tin" was used to pay taxes along the way. "Sealed tin" was subject to an export tax from Assur (wasitum: approx. 0.83 percent of the value of goods and donkeys), a transit tax between Assur and Kanesh (datum: 6–10 percent), an entry tax into Kanesh (nishatum: 3 percent), and an export tax from Kanesh (sadduatum: 6 percent).

[32] The lucrative potential of taxing caravan trade is highlighted in two letters from a client ruler in Syria, who complains that a change in Assyrian routes deprived him of toll income.

[33] For instance, merchants were not permitted to trade tin and textiles east of the Euphrates.

trade routes.

Assyrian merchants, however, did not always comply. In times of falling tin prices in Kanesh, some traders resorted to smuggling to evade tolls—despite the risks of imprisonment or fines. Assyrian texts reveal candid discussions among traders regarding smuggling routes, their potential profits, and associated risks.

7 Conclusion

This study provides the first econometric evidence suggesting that the discovery of metals and the associated emergence of long-distance trade played an important role in the rise of cities, states, and pronounced social hierarchies beginning in the fourth millennium BC.

Consistent with qualitative insights from the archaeological literature, we document that regions situated along trade corridors connecting copper and tin mines to fertile lands were significantly more likely to experience the Urban Revolution. We conjecture that geographic bottlenecks in these transit routes facilitated the emergence and persistence of tax-levying elites. We formally test this appropriability mechanism and support our findings with a series of historical case studies.

While our analysis centers on metal trade, we do not propose a monocausal explanation for the Urban Revolution. On the contrary, we acknowledge that additional factors—such as irrigation potential, storable cereals, and geographic circumscription—likely contributed to the emergence of complex societies.

Our analysis has two important limitations. First, no comprehensive dataset currently exists that consistently measures all archaeological markers of the Urban Revolution. Second, our empirical focus is limited to the Old World. We leave the investigation of how trade shaped early urbanization in the Americas to future research.

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